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# 1 Groups

## 1.1 Definitions and basic properties

**Definition 1.1.1** (Group). A *group* is a set  $G$  together with a binary operation  $(a, b) \mapsto a \cdot b = ab$  such that

- (i) (Associativity)  $(ab)c = a(bc)$  for all  $a, b, c \in G$ ;
- (ii) there exists an *identity element*  $1 \in G$  such that  $1 \cdot a = a \cdot 1 = a$  for all  $a \in G$ ;
- (iii) for all  $a \in G$ , there exists an *inverse*  $a^{-1} \in G$  of  $a$  such that  $aa^{-1} = a^{-1}a = 1$ .

Initially, it is not obvious that inverses are well-defined, as if the identity is not unique, then we are required to make an arbitrary choice in (iii). Furthermore, it may be that inverses are not unique, in which case the notation  $a^{-1}$  would not be well-defined. We resolve these concerns now.

**Proposition 1.1.2** (Uniqueness of identity / inverse). *Let  $G$  be a group.*

- 1. *There is exactly one identity element in  $G$ .*
- 2. *If  $a \in G$ , then  $a$  has exactly one inverse in  $G$ .*

*Proof.* 1. If  $1$  and  $1'$  are identities, then  $1' = 1 \cdot 1' = 1$ .

- 2. Let  $b$  and  $b'$  be inverses of  $a$ . Then  $b = 1 \cdot b = (b'a)b = b'(ab) = b' \cdot 1 = b'$ .

□

**Remark 1.1.3.** The associativity axiom allows us to define the product of any finitely many elements  $a_1 a_2 \dots a_n$ . For example,  $abcd = ((ab)c)d = (ab)(cd) = a(b(cd))$ .

**Proposition 1.1.4** (Cancellation). 1. *If  $ab = ac$ , then  $b = c$ .*

- 2. *If  $ac = bc$ , then  $a = b$ .*

**Notation.** For  $n \geq 0$ , write  $a^n$  for the  $n$ -fold product of  $a$  with itself.

**Proposition 1.1.5.** 1. *If  $ab = 1$  or  $ba = 1$ , then  $b = a^{-1}$ .*

- 2.  $(a^{-1})^{-1} = a$ .
- 3.  $(a_1 \dots a_n)^{-1} = a_n^{-1} \dots a_1^{-1}$ .
- 4.  $(a^{-1})^n = (a^n)^{-1}$  for  $n \geq 0$ . (Write  $a^{-n}$  for either of these expressions.)

**Proposition 1.1.6.** For  $n, m \in \mathbb{Z}$ ,  $a^n a^m = a^{n+m}$  and  $(a^n)^m = a^{nm}$ .

**Definition 1.1.7** (Order of a group element). Let  $a \in G$ . The *order* of  $a$ , denoted  $\text{ord } a$ , is the minimum  $n > 0$  such that  $a^n = 1$ . If such an  $n$  does not exist, then  $\text{ord } a = \infty$ .

**Definition 1.1.8** (Order of a group). The *order* of a group  $G$  is the cardinality  $|G|$  of  $G$  as a set. We say that  $G$  is *finite* if  $|G|$  is finite.

**Definition 1.1.9** (Abelian group). If  $G$  is a group and  $ab = ba$  for all  $a, b \in G$ , then  $G$  is said to be *abelian* (or *commutative*). The following notation, depending on context, may be used for an abelian group:

1. the binary operation is  $+$  instead of  $\cdot$ ;
2. the identity element is  $0$  instead of  $1$ ;
3. the inverse of  $a \in G$  is  $-a$  instead of  $a^{-1}$ .
4. the multiple  $na$  of  $a$  instead of  $n$ -th power  $a^n$ .

## 1.2 Examples of groups

**Example 1.2.1** (Trivial group). Any singleton set  $G = \{g\}$  can be made into an abelian group of order 1 with the operation  $gg = g$ . A generic trivial group may be written as  $1 = \{1\}$ .

If we are only working with abelian groups, then we may write  $0 = \{0\}$  for a generic trivial group.

**Example 1.2.2** (Numbers with addition). 1.  $(\mathbb{N}, +)$  is not a group, as  $1 \in \mathbb{N}$  has no inverse.  
2.  $(R, +)$  is an abelian group for  $R = \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ .

**Example 1.2.3** (Addition modulo  $n$ ). Let  $n$  be a positive integer. For  $a \in \mathbb{Z}$ , the *congruence class of  $a$  modulo  $n$*  is

$$[a]_n = \{a + nk \mid k \in \mathbb{Z}\} \subset \mathbb{Z}.$$

The relation  $\sim$  on  $\mathbb{Z}$  given by  $a \sim b \iff [a]_n = [b]_n$  is an equivalence relation whose equivalence classes are precisely the congruence classes modulo  $n$ , so these congruence classes partition  $\mathbb{Z}$ . The set of congruence classes modulo  $n$  is denoted  $\mathbb{Z}/n\mathbb{Z}$ , for reasons that will be seen later.

The operation  $[a]_n + [b]_n = [a + b]_n$  is well-defined and makes  $\mathbb{Z}/n\mathbb{Z}$  an abelian group of order  $n$ . For convenience, we may denote this additive group by  $\mathbb{Z}/n$ .

**Example 1.2.4** (Fields). If  $K$  is a field, then the set  $K^\times = K \setminus \{0\}$  with multiplication is an abelian group with identity 1. Familiar examples include  $\mathbb{Q}^\times$ ,  $\mathbb{R}^\times$ , and  $\mathbb{C}^\times$ .

With the language of groups, one could *define* a field to be a set  $K$  with distinct elements  $0, 1 \in K$  and operations  $+, \cdot$  such that

- (i) (addition)  $(K, +)$  is an abelian group with identity 0;
- (ii) (multiplication)  $(K^\times, \cdot)$  is an abelian group with identity 1;
- (iii) (distributive law)  $a(b + c) = ab + ac$  for all  $a, b, c \in K$ .

The first two axioms describe the separate structures of addition and multiplication, while the third axiom describes how they interact.

**Example 1.2.5** (Units modulo  $n$ ). The operation  $[a]_n \cdot [b]_n = [ab]_n$  is well-defined on  $\mathbb{Z}/n\mathbb{Z}$ , but it does not define a group structure (unless  $n = 1$ ). However, it does define a group structure on the subset  $(\mathbb{Z}/n\mathbb{Z})^\times = \{[a]_n \mid \gcd(a, n) = 1\}$ . This group is abelian of order  $\varphi(n)$ , where  $\varphi$  is the Euler totient function.

**Example 1.2.6** (Multiplication tables). Given a finite set  $G$ , we can define a group structure on  $G$  by writing down a full multiplication table which satisfies the group axioms. This is shown for the *Klein four-group*  $V_4$  (also denoted  $V$  or  $K_4$ ) below.

| $V_4$ | 1   | $a$ | $b$ | $c$ |
|-------|-----|-----|-----|-----|
| 1     | 1   | $a$ | $b$ | $c$ |
| $a$   | $a$ | 1   | $c$ | $b$ |
| $b$   | $b$ | $c$ | 1   | $a$ |
| $c$   | $c$ | $b$ | $a$ | 1   |

**Example 1.2.7** (Symmetric groups). Let  $X$  be any set. A *permutation* of  $X$  is a bijection from  $X$  to itself. The *symmetric group of  $X$* , denoted  $S(X)$ , is the group of permutations of  $X$  with function composition as the group operation.

If  $X$  is finite with  $|X| = n$  and the exact nature of the elements of  $X$  is not important, then we may assume  $X = \{1, \dots, n\}$ . In this case, we write  $S_n$  for  $S(X)$ . This is a group of order  $n!$ , and  $S_n$  is not abelian for  $n \geq 3$ .

**Example 1.2.8** (Matrix groups). Let  $F$  be a field. The set of invertible  $n \times n$  matrices with entries in  $F$ , together with matrix multiplication, form a group  $GL_n(F)$  called the *general linear group of degree  $n$  over  $F$* . This is not commutative for  $n \geq 2$ .

### 1.3 Homomorphisms

**Definition 1.3.1** (Group homomorphism). Let  $G$  and  $H$  be groups. A *(group) homomorphism* from  $G$  to  $H$  is a function  $f : G \rightarrow H$  such that  $f(ab) = f(a)f(b)$  for all  $a, b \in G$ .

If  $f : G \rightarrow H$  is a bijective homomorphism, then we say that  $f$  is an *isomorphism*.

Groups  $G$  and  $H$  are *isomorphic*, written  $G \cong H$ , if there exists an isomorphism  $f : G \rightarrow H$ .

**Proposition 1.3.2.** *Let  $f : G \rightarrow H$  be a homomorphism. Then*

1.  $f(1) = 1$ ;
2.  $f(a^{-1}) = f(a)^{-1}$  for all  $a \in G$ ;
3. if  $f$  is an isomorphism, then so is  $f^{-1} : H \rightarrow G$ .

*Proof.* 1.  $1 \cdot f(1) = f(1) = f(1 \cdot 1) = f(1) \cdot f(1)$ , so  $f(1) = 1$ .

$$2. f(a)f(a^{-1}) = f(aa^{-1}) = f(1) = 1.$$

3. The inverse of a bijective function is bijective, hence it is enough to show that  $f^{-1} : H \rightarrow G$  is a homomorphism. Since  $f$  is a homomorphism,

$$f(f^{-1}(ab)) = ab = f(f^{-1}(a))f(f^{-1}(b)) = f(f^{-1}(a)f^{-1}(b)).$$

As  $f$  is injective,  $f^{-1}(ab) = f^{-1}(a)f^{-1}(b)$ . □

**Proposition 1.3.3.** *If  $f : G \rightarrow H$  and  $g : H \rightarrow K$  are homomorphisms, then so is  $g \circ f : G \rightarrow K$ . If  $f$  and  $g$  are isomorphisms, then so is  $g \circ f$ .*

*Proof.* We have  $(g \circ f)(ab) = g(f(ab)) = g(f(a)f(b)) = g(f(a))g(f(b)) = (g \circ f)(a) \cdot (g \circ f)(b)$ .

If  $f, g$  are isomorphisms, then the inverse of  $g \circ f$  is  $f^{-1} \circ g^{-1}$ .  $\square$

**Example 1.3.4.** 1. Let  $G = \{g\}$  and  $H = \{h\}$  be trivial groups. The unique map  $f : G \rightarrow H$  with  $f(g) = h$  is an isomorphism, so there is only one trivial group up to isomorphism.

2. Two groups with the same multiplication tables, up to relabeling of elements, are isomorphic.

3. As additive groups,  $\mathbb{C} \cong \mathbb{R}^2$  with  $x + iy \leftrightarrow (x, y)$ .

4. The map  $x \mapsto e^x$  is a homomorphism  $\mathbb{R} \rightarrow \mathbb{R}^\times$ .

## 1.4 Cyclic groups

**Definition 1.4.1** (Generator / cyclic group). Let  $G$  be a group and  $a \in G$ . We say that  $a$  is a *generator of  $G$*  if every element of  $G$  is of the form  $a^n$  for some  $n \in \mathbb{Z}$ . If  $G$  has a generator, we say that  $G$  is *cyclic*.

**Example 1.4.2.** 1. The additive group  $\mathbb{Z}$  is an infinite cyclic group with generators  $\pm 1$ .

2. The additive group  $\mathbb{Z}/n\mathbb{Z}$  is a finite cyclic group. Its generators are  $[a]_n$  for  $\gcd(a, n) = 1$ . There are  $\varphi(n)$  such generators.

3. The multiplicative group  $(\mathbb{Z}/5\mathbb{Z})^\times$  is cyclic. The elements  $[2]_5$  and  $[3]_5$  are generators, while  $[1]_5$  and  $[4]_5$  are not.

**Theorem 1.4.3** (Classification of cyclic groups). *Every cyclic group is isomorphic to  $\mathbb{Z}$  or  $\mathbb{Z}/n\mathbb{Z}$  for some  $n > 0$ .*

*Proof.* Deferred (see Example 1.7.5).  $\square$

We write  $C_n$  for a cyclic group of order  $n$ . Infinite cyclic group is denoted by  $C_\infty$ .

## 1.5 Subgroups

**Definition 1.5.1** (Subgroup). Let  $G$  be a group and  $H \subset G$  be a subset. We say that  $H$  is a *subgroup of  $G$*  if it is a group with the operation inherited from  $G$ .

**Proposition 1.5.2.** *Let  $G$  be a group and  $H \subset G$  a subset. Then  $H$  is a subgroup of  $G$  if and only if*

(i)  $1 \in H$ ;

(ii) if  $a, b \in H$ , then  $ab \in H$ ;

(iii) if  $a \in H$ , then  $a^{-1} \in H$ .

**Corollary 1.5.3.** *Let  $G$  be a group and  $H \subset G$  a subset. Then  $H$  is a subgroup of  $G$  if and only if*

- (i)  $H$  is nonempty;
- (ii) if  $a, b \in H$ , then  $ab^{-1} \in H$ .

**Example 1.5.4.** 1. Every group  $G$  has the following subgroups  $\{1\} \subset G$  and  $G \subset G$ .

- 2. Every subgroup of  $\mathbb{Z}$  is of the form  $n\mathbb{Z} = \{nm \mid m \in \mathbb{Z}\}$  for  $n \geq 0$ .
- 3. As additive groups,  $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ .
- 4. As multiplicative groups,  $\mathbb{Q}^\times \subset \mathbb{R}^\times \subset \mathbb{C}^\times$ .
- 5. If  $\{H_i\}$  is a family of subgroups of  $G$ , then  $\bigcap_i H_i$  is a subgroup of  $G$ .
- 6. Let  $G$  be a group and  $a \in G$ . The *cyclic subgroup generated by  $a$*  is  $\langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$ . It is the smallest subgroup of  $G$  containing  $a$ .
- 7. Let  $G$  be a group and  $S \subset G$  be any subset. The *subgroup generated by  $S$*  is the smallest subgroup  $\langle S \rangle$  of  $G$  containing  $S$ . It is equivalently the subgroup of all finite products  $s_1 \cdots s_n$  with  $s_i \in S$  or  $s_i^{-1} \in S$  for each  $i$ .

**Definition 1.5.5** (Kernel / image). Let  $f : G \rightarrow H$  be a homomorphism.

- 1. The *kernel of  $f$*  is

$$\text{Ker } f = f^{-1}(1) = \{a \in G \mid f(a) = 1\} \subset G.$$

- 2. The *image of  $f$*  is

$$\text{Im } f = f(G) = \{f(a) \mid a \in G\} \subset H.$$

**Proposition 1.5.6.**  $\text{Ker } f \subset G$  and  $\text{Im } f \subset H$  are subgroups.

*Proof.* For  $\text{Ker } f \subset G$ , since  $f(1) = 1$ , we have  $1 \in \text{Ker } f$ . If  $a, b \in \text{Ker } f$ , then

$$f(ab^{-1}) = f(a)f(b)^{-1} = 1 \cdot 1^{-1} = 1,$$

so  $ab^{-1} \in \text{Ker } f$ . Thus  $\text{Ker } f \subset G$ .

For  $\text{Im } f \subset H$ , it is clear that  $\text{Im } f$  is non-empty. If  $x, y \in \text{Im } f$  with  $f(a) = x$  and  $f(b) = y$  for some  $a, b \in G$ , then

$$xy^{-1} = f(a)f(b)^{-1} = f(ab^{-1}) \in \text{Im } f,$$

so  $\text{Im } f \subset H$ . □

**Theorem 1.5.7.** *Let  $f : G \rightarrow H$  be a homomorphism. Then  $f$  is injective if and only if  $\text{Ker } f = 1$ .*

*Proof.* ( $\implies$ ) If  $f(g) = 1 = f(1)$ , then  $g = 1$ .

( $\impliedby$ ) Suppose  $\text{Ker } f = 1$  and  $f(a) = f(b)$  for  $a, b \in G$ . Then  $f(ab^{-1}) = 1$ , so  $ab^{-1} = 1$ . □

**Theorem 1.5.8.** *Let  $f : G \rightarrow H$  be an injective homomorphism. Then  $G \cong \text{Im } f \subset H$ .*

*Proof.* The homomorphism  $f : G \rightarrow \text{Im } f$  is injective and surjective.  $\square$

**Definition 1.5.9** (Embedding). If  $f : G \rightarrow H$  is injective, we say that  $f$  is an *embedding of  $G$  into  $H$* , written  $f : G \hookrightarrow H$ . That  $G$  embeds into  $H$  means that  $G$  is isomorphic to a subgroup of  $H$ .

**Example 1.5.10** (Cayley's theorem). Let  $G$  be a group. For  $a \in G$ , define the left multiplication function  $f_a : G \rightarrow G$  by  $f_a(g) = ag$ . This is not a homomorphism (unless  $a = 1$ ), but it does satisfy

$$f_a \circ f_b = f_{ab} \quad \text{and} \quad f_1 = 1_G.$$

In particular,  $f_a \circ f_{a^{-1}} = f_1 = 1_G$ , so each  $f_a$  is a bijection with  $(f_a)^{-1} = f_{a^{-1}}$ . Thus  $f_a \in S(G)$ , and the map  $G \hookrightarrow S(G)$  given by  $a \mapsto f_a$  is an injective homomorphism.

*Cayley's theorem* states that every group  $G$  embeds into some symmetric group. Our work here shows that in particular,  $G$  embeds into its own symmetric group  $S(G)$ .

**Definition 1.5.11** (Cosets). If  $H \subset G$  is a subgroup and  $a \in G$ , then  $aH = \{ah \mid h \in H\}$  is a *left coset of  $H$  in  $G$* , while  $Ha = \{ha \mid h \in H\}$  is a *right coset* (of  $H$  in  $G$ ).

Given  $H \subset G$ , say that  $a \sim b$  if  $b = ah$  for some  $h \in H$ , or equivalently, if  $a^{-1}b \in H$ . This is an equivalence relation, and the equivalence class of  $a$  is  $[a] = aH$ . Thus  $G$  is partitioned into left cosets of  $H$ . (These results can be developed similarly for right cosets.)

**Notation.** The set of left cosets of  $H$  in  $G$  is denoted  $G/H$ .

**Definition 1.5.12** (Index). The *index of  $H$  in  $G$*  is  $[G : H] = |G/H|$  (cardinality as a set).

**Theorem 1.5.13** (Lagrange). Let  $G$  be a group and  $H \subset G$ . Then  $|G| = [G : H] \cdot |H|$ .

*Proof.* Each coset  $X \in G/H$  has cardinality  $|H|$  and  $G = \bigsqcup_{X \in G/H} X$ .  $\square$

**Corollary 1.5.14.** Let  $G$  be a finite group.

1. If  $H \subset G$  is a subgroup, then  $|H|$  divides  $|G|$ .
2. If  $a \in G$ , then  $\text{ord } a$  divides  $|G|$ .

*Proof.* 1. Clear.

2. Apply the first statement to  $H = \langle a \rangle$ .  $\square$

## 1.6 Normal subgroups

**Definition 1.6.1** (Normal subgroup). A subgroup  $H \subset G$  is *normal* if  $aH = Ha$  for every  $a \in G$ . In this case we write  $H \triangleleft G$ .

**Example 1.6.2.** 1. In any group  $G$ , the subgroups  $\{1\}$  and  $G$  are normal.

2. In an abelian group  $G$ , every subgroup of  $G$  is normal.

**Proposition 1.6.3.**  $H \subset G$  is normal if and only if  $ghg^{-1} \in H$  for all  $g \in G$  and  $h \in H$ .

*Proof.* ( $\implies$ ) Let  $g \in G$  and  $h \in H$ . Since  $gH = Hg$ , we have  $gHg^{-1} = H$ .

( $\impliedby$ ) If  $ghg^{-1} \in H$  for all  $g \in G$  and  $h \in H$ , then  $gHg^{-1} \subset H$ , so  $gH \subset Hg$ . By the same reasoning applied to  $g^{-1}$  and  $h$ , we have  $Hg \subset gH$ , so  $gH = Hg$ .  $\square$

**Definition 1.6.4** (Conjugate). Given  $g, h \in G$ , the *conjugate of  $h$  by  $g$*  is  $ghg^{-1}$ .

**Proposition 1.6.5.** If  $N \triangleleft G$  and  $H \subset G$  is a subgroup with  $N \subset H$ , then  $N \triangleleft H$ .

**Proposition 1.6.6.** Let  $f : G \rightarrow H$  be a homomorphism. Then  $\text{Ker } f \triangleleft G$ .

*Proof.* Let  $k \in \text{Ker } f$  and  $g \in G$ . Then

$$f(gkg^{-1}) = f(g)f(k)f(g)^{-1} = f(g) \cdot 1 \cdot f(g)^{-1} = 1,$$

so  $gkg^{-1} \in \text{Ker } f$ .  $\square$

**Example 1.6.7.** Let  $F$  be a field and fix  $n \geq 1$ . Then  $\det : GL_n(F) \rightarrow F^\times$  is a homomorphism, so its kernel is a normal subgroup of  $GL_n(F)$ . The kernel of  $\det$  is the *special linear group*

$$SL_n(F) = \{A \in GL_n(F) \mid \det A = 1\}.$$

Let  $S$  and  $T$  be two subsets of a group  $G$ . We define the product  $ST$  as the subset of all elements in  $G$  of the form  $st$  for  $s \in S$  and  $t \in T$ .

**Proposition 1.6.8.** If  $H \triangleleft G$ , then the product of two cosets in  $G/H$  is a coset. Precisely,  $(aH)(bH) = abH$ . The product of cosets makes  $G/H$  a group.

*Proof.* This follows from the calculation

$$(aH)(bH) = aHbH = abHH = abH. \quad \square$$

**Definition 1.6.9** (Quotient group). If  $H \triangleleft G$ , then the group  $G/H$  is the *quotient group* or *factor group* of  $G$  by  $H$ . The map  $\pi : G \rightarrow G/H$  defined by  $\pi(a) = aH$  is the *canonical homomorphism* or *quotient homomorphism*.

Note that  $\text{Ker } \pi = H$  and  $\text{Im } \pi = G/H$ . Thus *every* normal subgroup of  $G$  is the kernel of some homomorphism from  $G$  to another group.

**Example 1.6.10.** 1. For any group  $G$ , we have  $G/G \cong \{1\}$  and  $G/\{1\} \cong G$ .

2. The subgroup  $n\mathbb{Z} \subset \mathbb{Z}$  is normal since  $\mathbb{Z}$  is abelian, and  $\mathbb{Z}/n\mathbb{Z}$  is the additive group of integers modulo  $n$ , in accordance with our earlier use of the notation  $\mathbb{Z}/n\mathbb{Z}$ .

3. The elements of  $\mathbb{C}/\mathbb{R}$  are lines  $l_y = \{x + iy \mid y \in \mathbb{R}\}$ . This is isomorphic to  $\mathbb{R}$  via  $l_y \mapsto y$ .

**Theorem 1.6.11** (Correspondence theorem). Let  $H \triangleleft G$ . There is a natural bijection

$$\{\text{subgroups of } G \text{ containing } H\} \longleftrightarrow \{\text{subgroups of } G/H\}$$

$$K \longmapsto \pi(K)$$

$$\pi^{-1}(L) \rightarrow L.$$

Furthermore, normal subgroups of  $G$  containing  $H$  are paired with normal subgroups of  $G/H$ .  $\square$

## 1.7 Isomorphism theorems

**Definition 1.7.1** (Factoring through). Let  $f : G \rightarrow H$  be a homomorphism and  $N \triangleleft G$  with the canonical homomorphism  $\pi : G \rightarrow G/N$ . Then  $f$  *factors through*  $G/N$  if there is a homomorphism  $\bar{f} : G/N \rightarrow H$  such that  $f = \bar{f} \circ \pi$ .

**Theorem 1.7.2.** *Let  $f : G \rightarrow H$  be a homomorphism and  $N \triangleleft G$ . Then  $f$  factors uniquely through  $G/N$  if and only if  $N \subset \text{Ker } f$ .*

*Proof.* ( $\implies$ ) Suppose  $f$  factors through  $G/N$  as  $f = \bar{f} \circ \pi$ . Then

$$f(N) = \bar{f}(\pi(N)) = \bar{f}(1) = 1,$$

so  $N \subset \text{Ker } f$ .

( $\impliedby$ ) Suppose  $N \subset \text{Ker } f$ . For  $f$  to factor as  $\bar{f} \circ \pi$ , we must have

$$\bar{f}(aN) = (\bar{f} \circ \pi)(a) = f(a),$$

so we take this to define  $\bar{f}$  and show that  $\bar{f}$  is well-defined. If  $aN = bN$ , then  $a^{-1}b \in N \subset \text{Ker } f$ , so  $f(a^{-1}b) = 1$ , hence

$$\bar{f}(aN) = f(a) = f(b) = \bar{f}(bN).$$

The proof that  $\bar{f}$  is a homomorphism is omitted. □

**Theorem 1.7.3** (First isomorphism theorem). *Let  $f : G \rightarrow H$  be a group homomorphism. Then  $G/\text{Ker } f \cong \text{Im } f$ , with isomorphism  $\bar{f}$  given by factoring  $f$  through  $G/\text{Ker } f$ . Precisely,  $\bar{f}(a\text{Ker } f) = f(a)$ .*

*Proof.* Let  $\pi : G \rightarrow G/\text{Ker } f$  be the canonical homomorphism, so  $f = \bar{f} \circ \pi$ . Then  $\text{Im } f = \text{Im } \bar{f}$ , so  $\bar{f} : G/\text{Ker } f \rightarrow \text{Im } f$  is surjective. To see that it is injective, suppose  $a\text{Ker } f \in \text{Ker } \bar{f}$ . Then

$$1 = \bar{f}(a\text{Ker } f) = (\bar{f} \circ \pi)(a) = f(a),$$

so  $a \in \text{Ker } f$  and  $a\text{Ker } f = \text{Ker } f$  is the identity in  $G/\text{Ker } f$ . □

**Corollary 1.7.4.** *If  $f : G \rightarrow H$  is a surjective homomorphism, then  $G/\text{Ker } f \cong H$ .*

**Example 1.7.5.** We prove Theorem 1.4.3 on the classification of cyclic groups.

Let  $G$  be a cyclic group generated by  $a$ , and define the homomorphism  $f : \mathbb{Z} \rightarrow G$  by  $n \mapsto a^n$ . This is surjective since  $G$  is cyclic, so  $\mathbb{Z}/\text{Ker } f \cong G$ . Since  $\text{Ker } f \subset \mathbb{Z}$ , it is of the form  $n\mathbb{Z}$  for some  $n \geq 0$ .

If  $n = 0$ , then  $G \cong \mathbb{Z}$ . Otherwise,  $G \cong \mathbb{Z}/n\mathbb{Z}$ .

**Theorem 1.7.6** (Second isomorphism theorem). *Let  $K \subset G$  be a subgroup and  $N \triangleleft G$ . Then*

1.  $KN \subset G$ ;
2.  $N \triangleleft KN$ ;

3.  $K \cap N \triangleleft K$ ;
4.  $KN/N \cong K/(K \cap N)$ .

*Proof.* Here  $KN$  is non-empty, and

$$(KN)(KN)^{-1} = KNNK = KNNK = KKN = KN,$$

hence  $KN$  is a subgroup in  $G$  containing  $N$ . Since  $N \triangleleft G$  we have  $N \triangleleft KN$ .

Define a homomorphism  $f : K \rightarrow KN/N$  by  $k \mapsto kN$ . It is surjective since  $knN = kN \operatorname{Im}(f)$  and  $\operatorname{Ker}(f) = K \cap N$ . It follows that  $K \cap N \triangleleft K$  and  $K/K \cap N \cong KN/N$  by the first isomorphism theorem.  $\square$

**Theorem 1.7.7** (Third isomorphism theorem). *Let  $K, H \triangleleft G$  with  $K \subset H$ . Then*

1.  $H/K \triangleleft G/K$ ;
2.  $(G/K)/(H/K) \cong G/H$ .

*Proof.* Define  $f : G/K \rightarrow G/H$  by  $gK \mapsto gH$ . This is a well-defined surjective homomorphism with kernel  $H/K$  (therefore normal in  $G/K$ ), hence  $(G/K)/(H/K) \cong G/H$  by the first isomorphism theorem.  $\square$

## 1.8 Group actions

**Definition 1.8.1** (Group action). Let  $G$  be a group and  $X$  be a set. An *action of  $G$  on  $X$*  is a map

$$\begin{aligned} \theta : G \times X &\longrightarrow X \\ (g, x) &\rightarrow gx \end{aligned}$$

such that

- (i)  $1x = x$  for all  $x \in X$ ;
- (ii)  $g(hx) = (gh)x$  for all  $g, h \in G$  and  $x \in X$ .

**Example 1.8.2.** Let  $X$  be a set. Then  $S(X)$  acts on  $X$  by  $(f, x) \mapsto f(x)$  for  $f \in S(X)$  and  $x \in X$ .

**Definition 1.8.3** (Pullback of a group action). Let  $f : G \rightarrow H$  be a homomorphism and suppose  $\theta$  is an action of  $H$  on  $X$ . The *pullback of  $\theta$  by  $f$*  is the action of  $G$  on  $X$  given by  $gx = f(g)x$ .

There is a bijection

$$\{\text{actions of } G \text{ on } X\} \longleftrightarrow \{\text{homomorphisms } G \rightarrow S(X)\}.$$

Indeed, a  $G$ -action on  $X$  yields a group homomorphism  $f : G \rightarrow S(X)$  by  $f(g)(x) = gx$ . Conversely, a homomorphism  $f$  defines a  $G$ -action on  $X$  via  $gx = f(g)(x)$ .

This tells us that every group action is the pullback of the action in Example 1.8.2 for some set  $X$ . Thus this action may be called the *universal action*.

**Definition 1.8.4** (Kernel / faithful action). Let  $\theta$  be an action of  $G$  on  $X$ . The *kernel* of  $\theta$  is the kernel of the induced homomorphism  $G \rightarrow S(X)$ . We say that  $\theta$  is *faithful* if  $\text{Ker } \theta = 1$ .

**Definition 1.8.5** (Orbit / stabilizer). Let  $G$  act on  $X$  and let  $x \in X$ .

1. The *orbit* of  $x$ , written  $\text{orb } x$  or  $Gx$ , is

$$\text{orb } x = \{gx \mid g \in G\} \subset X.$$

2. The *stabilizer* of  $x$ , written  $\text{stab } x$  or  $G_x$ , is

$$\text{stab } x = \{g \in G \mid gx = x\} \subset G.$$

**Proposition 1.8.6.**  $\text{Ker } \theta = \bigcap_x \text{stab } x$ .

Define a relation on  $X$  by  $x \sim y$  if  $\text{orb } x = \text{orb } y$ . Then  $\sim$  is an equivalence relation whose equivalence classes are the orbits of  $\theta$ . In particular, the orbits partition  $X$ .

**Definition 1.8.7** (Transitive action). An action of  $G$  on  $X$  is *transitive* if the only orbit is  $X$ .

**Example 1.8.8** (Trivial action). The *trivial action* of  $G$  on  $X$  is given by  $gx = x$  for all  $g \in G$  and  $x \in X$ . This is not faithful (unless  $G = 1$ ) and not transitive (unless  $|X| = 1$ ). We have  $\text{orb } x = \{x\}$  and  $\text{stab } x = G$ .

**Example 1.8.9** (Regular action). The *regular action* of  $G$  on  $X = G$  is the action given by  $(g, x) \mapsto gx$ . This is faithful and transitive, as  $\text{stab } x = 1$  and  $\text{orb } x = G$  for all  $x \in G$ . The induced homomorphism  $G \rightarrow S(G)$  is the embedding from Example 1.5.10.

**Example 1.8.10** (Left coset action). If  $H \subset G$ , then the *left coset action* of  $G$  on  $X = G/H$  is the transitive action given by  $(g, xH) \mapsto gxH$ . Given  $gH \in G/H$ , we have  $\text{orb}(gH) = G/H$  and  $\text{stab}(gH) = gHg^{-1}$ . The kernel of the left coset action is the *normal core* of  $H$  in  $G$ , which is the largest normal subgroup of  $G$  contained in  $H$ .

**Definition 1.8.11** (Automorphism). An *automorphism* of  $G$  is an isomorphism  $G \rightarrow G$ . The group of automorphisms of  $G$  is denoted  $\text{Aut } G$ .

**Example 1.8.12.** 1.  $\text{Aut } \mathbb{Z} = \{(n \mapsto n), (n \mapsto -n)\}$ .

2.  $\text{Aut}(\mathbb{Z}/n\mathbb{Z}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$ .

**Example 1.8.13.** Any automorphism of  $G$  is also a permutation of  $G$ , so  $\text{Aut}(G) \subset S(G)$ . Then  $\text{Aut } G$  acts on  $G$  as the pullback of the universal action by the inclusion  $\text{Aut}(G) \hookrightarrow S(G)$ .

**Example 1.8.14** (Conjugation action on group elements). Let  $G$  act on  $X = G$  by conjugation, i.e.  $(g, x) \mapsto gxg^{-1}$ . The orbit of  $x \in G$  is the *conjugacy class* of  $x$  in  $G$ , while the stabilizer of  $x$  is the *centralizer* of  $x$  in  $G$ . For each  $g \in G$ , the map  $x \mapsto gxg^{-1}$  is an automorphism of  $G$ .

**Definition 1.8.15** (Center). The *center* of  $G$ , denoted  $Z$  or  $Z(G)$ , is the kernel of the conjugation action on the elements of  $G$ . Equivalently,  $Z = \{g \in G \mid gh = hg \text{ for all } h \in G\}$ . In particular,  $Z = G$  if and only if  $G$  is abelian.

**Definition 1.8.16** (Inner automorphism). An *inner automorphism* of  $G$  is an automorphism of the form  $x \mapsto gxg^{-1}$  for some  $g \in G$ . The group of inner automorphisms is denoted  $\text{Inn } G$ , and is a subgroup of  $\text{Aut } G$ .

**Proposition 1.8.17.**  $\text{Inn } G \cong G/Z$ .

*Proof.* Apply the first isomorphism theorem to the homomorphism induced by the conjugation action on the elements of  $G$ .  $\square$

**Example 1.8.18** (Conjugation action on subgroups). Let  $G$  act on the set  $X$  of all subgroups of  $G$  by conjugation, i.e.  $(g, H) \mapsto gHg^{-1}$ . The stabilizer of  $H$  is the *normalizer* of  $H$  in  $G$ , denoted  $N_G(H)$ . It is the largest subgroup of  $G$  in which  $H$  is normal.

**Lemma 1.8.19.** Let  $G$  be a finite group and  $H \subset G$  such that  $[G : H]$  is the smallest prime divisor of  $|G|$ . Then  $H \triangleleft G$ .

*Proof.* Let  $p$  be the smallest prime divisor of  $|G|$ . Let  $G$  act on  $G/H$  by left translation and  $f : G \rightarrow S(X) \cong S_p$  be the induced homomorphism. Let  $N = \text{Ker } f \triangleleft G$ . Then  $N \subset H$  and  $|G/N|$  divides  $p!$  by the first isomorphism theorem, so in particular  $|G/N|$  has no prime factor greater than  $p$ . On the other hand,  $|G/N|$  divides  $|G|$ , and  $|G|$  has no prime factor less than  $p$ . Thus  $|G/N| = p$ , i.e.  $[G : N] = p$ . Since  $G$  is finite, this means that  $H = N \triangleleft G$ .  $\square$

**Theorem 1.8.20** (Orbit-stabilizer). Let  $G$  be a group acting on a set  $X$  and let  $x \in X$ . Then  $|\text{orb } x| = [G : \text{stab } x]$ . In particular, if  $G$  is finite, then  $|\text{orb } x| = |G|/|\text{stab } x|$ .

*Proof.* Let  $y \in \text{orb } x$ . Then there exists  $g_y$  such that  $g_y x = y$ . Define a function  $\text{orb } x \rightarrow G/\text{stab } x$  by  $y \mapsto g_y \text{stab } x$ . This is well-defined, as if  $g$  satisfies  $gx = y$ , then  $g^{-1}g_y x = x$ , so  $g^{-1}g_y \in \text{stab } x$  and  $g_y \text{stab } x = g \text{stab } x$ . The inverse of this function is  $g \text{stab } x \mapsto gx$ . Thus we have a bijection between  $\text{orb } x$  and  $G/\text{stab } x$ .  $\square$

**Example 1.8.21.** If the group  $G$  is finite and  $H \subset G$ , then the number of subgroups conjugate to  $H$  is  $|G|/|N_G(H)|$ .

**Definition 1.8.22** (Fixed point). Given an action of  $G$  on  $X$  and  $S \subset G$ , a *fixed point* of  $g$  is an element  $x \in X$  with  $gx = x$  for all  $g \in S$ . The set of fixed points of  $S$  is denoted  $X^S$ . When  $S = \{g\}$ , we write  $X^g$  for  $X^S$ .

**Proposition 1.8.23.** If  $H = \langle S \rangle \subset G$ , then  $X^S = X^H$ .

**Lemma 1.8.24** (Burnside). Let  $G$  be a finite group acting on a finite set  $X$ . Then the number of orbits of the action is

$$\frac{1}{|G|} \sum_{g \in G} |X^g|.$$

*Proof.* The number of orbits can be counted as a weighted sum over all  $x \in X$  by  $1/|\text{orb } x|$ , so we get by orbit-stabilizer

$$\begin{aligned} \sum_{x \in X} \frac{1}{|\text{orb } x|} &= \sum_{x \in X} \frac{|\text{stab } x|}{|G|} = \frac{1}{|G|} \sum_{x \in X} |\text{stab } x| = \frac{1}{|G|} |\{(g, x) \in G \times X \mid gx = x\}| \\ &= \frac{1}{|G|} \sum_{g \in G} |\{x \in X \mid gx = x\}| = \frac{1}{|G|} \sum_{g \in G} |X^g|. \end{aligned} \quad \square$$

## 1.9 Sylow theorems

Throughout, let  $G$  be a finite group and  $p$  be a prime.

**Definition 1.9.1** ( $p$ -group). 1.  $G$  is a  $p$ -group if  $|G| = p^n$  for some  $n \geq 0$ .

2. If  $H \subset G$ , then  $H$  is a  $p$ -subgroup of  $G$  if  $H$  is a  $p$ -group. (Here  $G$  need not be a  $p$ -group.)

**Lemma 1.9.2.** Let a  $p$ -group  $H$  act on a finite set  $X$ . Then  $|X^H| \equiv |X| \pmod{p}$ .

*Proof.* If  $X^H = \{x_1, \dots, x_k\}$ , then  $\text{orb}(x_i) = \{x_i\}$  for each  $i$ . All other orbits have size divisible by  $p$  by orbit-stabilizer, hence the result.  $\square$

**Theorem 1.9.3** (Cauchy). If  $p$  divides  $|G|$ , then  $G$  has an element of order  $p$ .

*Proof.* Let  $X = \{(x_1, \dots, x_p) \in G^p \mid x_1 \cdots x_p = 1\}$ . Then  $|X| = |G|^{p-1}$ , which is divisible by  $p$ . Consider the action of a cyclic group  $H$  of order  $p$  with a generator  $\sigma$  on  $X$  defined by  $\sigma(x_1, \dots, x_p) = (x_p, x_1, \dots, x_{p-1})$ . Since  $|H| = p$ , it is a  $p$ -group, so by Lemma 1.9.2,  $|X^H| \equiv |X| \equiv 0 \pmod{p}$ . Any fixed point (element of  $X^H$ ) is of the form  $(x, \dots, x)$  with  $x^p = 1$ . Note that  $(1, \dots, 1) \in X^H$ , so  $|X^H| \geq 1$  and is divisible by  $p \geq 2$ . Thus there exists  $(x, \dots, x) \in X^H$  with  $x \neq 1$ , so  $x$  is an element of order  $p$ .  $\square$

**Theorem 1.9.4.** Let  $G$  be a non-trivial  $p$ -group. Then  $Z(G) \neq 1$ .

*Proof.* Let  $G$  act on  $X = G$  by conjugation, so  $X^G = Z(G)$ . By Lemma 1.9.2,  $|Z(G)| \equiv |G| \equiv 0 \pmod{p}$ , so  $Z(G)$  is non-trivial.  $\square$

**Lemma 1.9.5.** Let  $H$  be a  $p$ -subgroup of a finite group  $G$ . Then  $[N_G(H) : H] \equiv [G : H] \pmod{p}$ .

*Proof.* Let  $H$  act on  $X = G/H$  by left multiplication. If  $gH \in X^H$ , i.e.  $h(gH) = gH$  for all  $h \in H$ , then  $g \in N_G(H)$ . Hence  $|X^H| = [N_G(H) : H]$ . The result follows by Lemma 1.9.2.  $\square$

**Definition 1.9.6** (Sylow  $p$ -subgroup). Write  $|G| = p^n m$  with  $n > 0$  and  $p \nmid m$ . A Sylow  $p$ -subgroup  $H \subset G$  is a subgroup of  $G$  with  $|H| = p^n$ .

**Theorem 1.9.7** (First Sylow theorem). Let  $p$  be a prime and  $G$  be a finite group with  $p$  dividing  $|G|$ . If  $H \subset G$  is a  $p$ -subgroup, then

1. If  $H$  is not a Sylow  $p$ -subgroup, then  $H$  lies in a subgroup  $N$  of order  $p \cdot |H|$  with  $H \triangleleft N$ ;
2.  $H$  is contained in a Sylow  $p$ -subgroup of  $G$ .

*Proof.* 1. Let  $|H| = p^i$  for some  $i < n$ . Then  $[N_G(H) : H] \equiv [G : H] = p^{n-i} m \equiv 0 \pmod{p}$  by Lemma 1.9.5, so  $N_G(H)/H$  has order divisible by  $p$ . By Cauchy's theorem, it has an element of order  $p$ , hence a subgroup  $F$  of order  $p$ . By the correspondence theorem, if  $\pi : N_G(H) \rightarrow N_G(H)/H$  is the canonical homomorphism, then  $N = \pi^{-1}(F)$  is a subgroup of  $N_G(H)$  containing  $H$  and  $[N : H] = p$ . Since  $H \triangleleft N_G(H)$ , we also have  $H \triangleleft N$ .

2. Apply the first statement repeatedly.  $\square$

**Theorem 1.9.8** (Second Sylow theorem). Suppose  $p \mid |G|$ .

1. If  $H \subset G$  is a  $p$ -subgroup and  $P \subset G$  is a Sylow  $p$ -subgroup, then  $gHg^{-1} \subset P$  for some  $g \in G$ .
2. Any two Sylow  $p$ -subgroups of  $G$  are conjugate.

*Proof.* 1. Let  $H$  act on  $X = G/P$  by left multiplication. Then  $|X^H| \equiv |X| = m \not\equiv 0 \pmod{p}$ , so  $|X^H| \geq 1$ . Let  $gP \in X^H$ . Then  $hgP = gP$  for all  $h \in H$ , so  $g^{-1}Hg \subset P$ .

2. Immediate from the first statement.  $\square$

**Corollary 1.9.9.** *Let  $P \subset G$  be a Sylow  $p$ -subgroup. Then  $P \triangleleft G$  if and only if  $P$  is the unique Sylow  $p$ -subgroup of  $G$ .*

**Notation.** Given a finite group  $G$  and prime  $p$ , write  $\text{Syl}_p(G)$  for the set of all Sylow  $p$ -subgroups of  $G$  and  $n_p = |\text{Syl}_p(G)|$ .

**Theorem 1.9.10** (Third Sylow theorem). *Suppose  $|G| = p^n m$  with  $n > 0$  and  $p \nmid m$ . Then  $n_p \mid m$  and  $n_p \equiv 1 \pmod{p}$ .*

*Proof.* By the second Sylow theorem,  $G$  acts transitively on  $X = \text{Syl}_p(G)$  by conjugation. Let  $P \in X$  be some Sylow  $p$ -subgroup. Then  $\text{orb } P = X$  and  $\text{stab } P = N_G(P)$ , so by orbit-stabilizer,  $n_p = |X| = [G : N_G(P)]$ . This divides  $[G : P] = m$ .

Now let  $P$  act on  $X$  by conjugation. Since  $P$  is a  $p$ -group,  $|X^P| \equiv |X| = n_p \pmod{p}$  by Lemma 1.9.2. If  $Q \in X^P$ , then  $gQg^{-1} = Q$  for all  $g \in P$ . Thus  $P, Q \subset N_G(Q)$  are Sylow  $p$ -subgroups. Since  $Q \triangleleft N_G(Q)$ , this means that  $Q = P$ . Thus  $X^P = \{P\}$  and  $n_p \equiv 1 \pmod{p}$ .  $\square$

## 1.10 Direct products

**Definition 1.10.1** (Direct product). Let  $G_1, \dots, G_n$  be groups. Then  $G = G_1 \times \dots \times G_n$  forms a group with componentwise group operations, called the *(external) direct product* of  $G_1, \dots, G_n$ .

**Example 1.10.2.** 1.  $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$  has order 4 and is non-cyclic, so it is isomorphic to  $V_4$ .

2. The direct product of abelian groups is abelian.

**Definition 1.10.3** (Internal direct product). Let  $H_1, \dots, H_n \subset G$ . We say that  $G$  is the *internal direct product* of  $H_1, \dots, H_n$  if the multiplication map

$$\begin{aligned} f : H_1 \times \dots \times H_n &\longrightarrow G \\ (h_1, \dots, h_n) &\longmapsto h_1 \dots h_n \end{aligned}$$

is a group isomorphism.

**Proposition 1.10.4.** *If  $G$  is the internal direct product of  $H_1, \dots, H_n$ , then  $G \cong H_1 \times \dots \times H_n$ .*

**Theorem 1.10.5** (Direct product theorem). *Let  $G$  be a group and  $H_1, \dots, H_n \subset G$  be subgroups. Then  $G$  is the internal direct product of  $H_1, \dots, H_n$  if and only if*

1.  $H_i \triangleleft G$ ;
2.  $G = H_1 \dots H_n$ ;

3. if  $1 = h_1 \cdots h_n$  for  $h_i \in H_i$ , then  $h_i = 1$  for all  $i$ .

*Proof.* ( $\implies$ ) Let  $f$  be the multiplication isomorphism above. The first condition follows from the fact that  $f^{-1}(H_i)$  is the copy of  $H_i$  embedded in  $H_1 \times \cdots \times H_n$  in the natural way, which is normal. The last two conditions are surjectivity and injectivity of  $f$ , respectively.

( $\impliedby$ ) By the third condition,  $H_i \cap H_j = 1$  for  $i \neq j$ . Then if  $h_i \in H_i$  and  $h_j \in H_j$ , we have  $h_i h_j h_i^{-1} h_j^{-1} \in H_i \cap H_j = 1$  by normality of  $H_i$  and  $H_j$  in  $G$ , so  $H_i$  and  $H_j$  commute. This is enough to ensure that the multiplication map  $f$  above is a group homomorphism. That  $f$  is surjective and injective follows from the second and third conditions, respectively.  $\square$

**Corollary 1.10.6.** 1. If  $n = 2$ , then condition 3 may be replaced by  $H_1 \cap H_2 = 1$ .

2. If  $G$  is finite, then one of conditions 2 or 3 may be replaced by  $|G| = |H_1| \cdots |H_n|$ .

**Example 1.10.7** (Chinese remainder theorem). Let  $m$  and  $n$  be relatively prime positive integers, and consider  $\mathbb{Z}/mn\mathbb{Z}$ . The subgroups  $m\mathbb{Z}/mn\mathbb{Z}$  and  $n\mathbb{Z}/mn\mathbb{Z}$  satisfy the conditions of the direct product theorem, so

$$\mathbb{Z}/mn\mathbb{Z} \cong (m\mathbb{Z}/mn\mathbb{Z}) \times (n\mathbb{Z}/mn\mathbb{Z}) \cong (\mathbb{Z}/n\mathbb{Z}) \times (\mathbb{Z}/m\mathbb{Z}).$$

**Theorem 1.10.8.** Let  $G$  be a finite group with  $|G| = p_1^{k_1} \cdots p_n^{k_n}$ , where the  $p_i$  are distinct primes. For each  $i$ , let  $H_i \subset G$  be a Sylow  $p_i$ -subgroup. If  $H_i \triangleleft G$  for all  $i$ , then  $G$  is the internal direct product of  $H_1, \dots, H_n$ .

*Proof.* We apply the direct product theorem.

1. This is given.
2. We replace  $G = H_1 \cdots H_n$  with  $|G| = |H_1| \cdots |H_n|$ , and this holds since  $|H_i| = p_i^{k_i}$  for each  $i$ .
3. Suppose  $1 = h_1 \cdots h_n = 1$  with  $h_i \in H_i$  for each  $i$ . Since each  $H_i$  is normal in  $G$ , the product  $H_1 \cdots H_{n-1}$  is a subgroup of  $G$  of order not divisible by  $p_n$ , so  $h_1 \cdots h_{n-1} = h_n^{-1}$  has order not divisible by  $p_n$ . On the other hand,  $h_n \in H_n$ , so  $h_n$  has order 1 or order divisible by  $p_n$ . Thus we must have  $h_n = 1$ . By the same reasoning,  $h_i = 1$  for all  $i$ , as required.  $\square$

**Proposition 1.10.9.** Let  $G$  be a group of order  $pq$ , where  $p < q$  are primes. If  $q \not\equiv 1 \pmod{p}$ , then  $G$  is cyclic.

*Proof.* If  $H_p$  and  $H_q$  are Sylow subgroups, then  $|H_p| = p$  and  $|H_q| = q$ . By Sylow's third theorem,  $H_p$  and  $H_q$  are normal (here we use  $q \not\equiv 1 \pmod{p}$ ). Thus

$$G \cong H_p \times H_q \cong (\mathbb{Z}/p\mathbb{Z}) \times (\mathbb{Z}/q\mathbb{Z}) \cong \mathbb{Z}/pq\mathbb{Z}$$

by the Chinese remainder theorem.  $\square$

### 1.11 Nilpotent and solvable groups

**Definition 1.11.1** (Commutator / commutator subgroup). Given  $g, h \in G$ , the *commutator* of  $g$  and  $h$  is  $[g, h] = ghg^{-1}h^{-1}$ .

If  $H, K \subset G$ , then the *commutator subgroup*  $[H, K]$  of  $G$  is the subgroup generated by commutators  $[h, k]$  with  $h \in H$  and  $k \in K$ .

**Proposition 1.11.2.** 1.  $[g, h]^{-1} = [h, g]$ ;

2.  $g[h, k]g^{-1} = [ghg^{-1}, gkg^{-1}]$ ;

3.  $[x, y] = 1$  if and only if  $xy = yx$ .

**Corollary 1.11.3.**  $[G, G] = 1$  if and only if  $G$  is abelian.

**Proposition 1.11.4.** 1. If  $H, K \triangleleft G$ , then  $[H, K] \triangleleft G$ .

2. If  $H \subset G$ , then  $[G, H] \triangleleft G$ .

3. If  $H, K \subset G$  and  $[H, K] \subset H$ , then  $K \subset N_G(H)$ .

*Proof.* 2) Let  $x, g \in G$  and  $h \in H$ . Then

$$x[g, h]x^{-1} = xghg^{-1}h^{-1}x^{-1} = xghg^{-1}x^{-1}h^{-1}hxx^{-1} = [xg, h] \cdot [x, h]^{-1} \quad \square$$

**Definition 1.11.5** (Central series / lower central series). A *central series* is a series of subgroups

$$G = G_0 \triangleright G_1 \triangleright \cdots \triangleright G_n = 1$$

such that  $G_i \triangleleft G$  and  $[G, G_i] \subset G_{i+1}$  for each  $i$ , or equivalently,  $G_i/G_{i+1} \subset Z(G/G_{i+1})$  for each  $i$ .

The *lower central series* is the series

$$G = G_0 \triangleright G_1 \triangleright \cdots$$

with  $G_{i+1} = [G, G_i]$  for each  $i$ .

**Definition 1.11.6** (Nilpotent group). A group is *nilpotent* if it has a (terminating) central series. Equivalently, the lower central series terminates in the trivial group.

**Example 1.11.7.** 1. Abelian groups are nilpotent.

2. Products of finitely many nilpotent groups are nilpotent.

**Lemma 1.11.8.** If  $G/Z$  is nilpotent, then  $G$  is nilpotent.

**Corollary 1.11.9.** Every  $p$ -group is nilpotent.

**Lemma 1.11.10.** Let  $G$  be a nilpotent group and  $H \subset G$  be a proper subgroup. Then  $H \neq N_G(H)$ .

*Proof.* Take a central series  $G = G_0 \triangleright G_1 \triangleright \cdots \triangleright G_n = 1$ . There exists  $j$  such that  $G_j \not\subset H$  but  $G_{j+1} \subset H$ . Then  $[H, G_j] \subset [G, G_j] \subset G_{j+1} \subset H$ , so  $G_j \subset N_G(H)$ , which means  $N_G(H) \neq H$ .  $\square$

**Lemma 1.11.11.** Let  $P$  be a Sylow  $p$ -subgroup of  $G$  and  $H = N_G(P)$ . Then  $N_G(H) = H$ .

*Proof.* Let  $g \in N_G(H)$ . Then  $P$  and  $gPg^{-1}$  are Sylow  $p$ -subgroups of  $H$ . Since  $P$  is normal in  $H$ , we have  $gPg^{-1} = P$ , so  $g \in H$ .  $\square$

**Theorem 1.11.12.** *A finite group  $G$  is nilpotent if and only if  $G$  is a product of  $p$ -groups.*

*Proof.* ( $\Leftarrow$ ) Products of nilpotent groups, in particular  $p$ -groups, are nilpotent.

( $\Rightarrow$ ) By Theorem 1.10.8, it suffices to show that every Sylow  $p$ -subgroup  $P \subset G$  is normal. Let  $H = N_G(P)$ . Then  $N_G(H) = H$  by Lemma 1.11.11, so by Lemma 1.11.10,  $H$  cannot be a proper subgroup of  $G$ , i.e.  $H = G$ . Thus  $P \triangleleft G$ .  $\square$

**Proposition 1.11.13.** *Let  $N \triangleleft G$ . Then  $G/N$  is abelian if and only if  $[G, G] \subset N$ .*

*Proof.* ( $\Rightarrow$ ) Let  $g, h \in G$ . Then since  $G/N$  is abelian,  $[gN, hN] = [g, h]N = N$ , so  $[g, h] \in N$ . Since  $N$  contains all commutators, it contains  $[G, G]$ .

( $\Leftarrow$ ) If  $[G, G] \subset N$ , then  $G/N \cong (G/[G, G])/(N/[G, G])$ , so it suffices to show that  $G/[G, G]$  is abelian. This is immediate from  $[G, G]$  containing all commutators.  $\square$

**Definition 1.11.14** (Derived subgroup / abelianization). The *derived subgroup* of  $G$  is  $G' = [G, G]$ . The *abelianization* of  $G$  is  $G/G'$ .

**Definition 1.11.15** (Derived series). The *derived series* of  $G$  is

$$G = G^{(0)} \triangleright G^{(1)} \triangleright \dots$$

with  $G^{(i+1)} = [G^{(i)}, G^{(i)}]$  for each  $i$ .

**Definition 1.11.16** (Solvable group). We say that  $G$  is *solvable* if its derived series terminates in the trivial group. Equivalently, there is a sequence of subgroups

$$G = G_0 \triangleright G_1 \triangleright \dots \triangleright G_n = 1$$

with  $G_i/G_{i+1}$  abelian for each  $i$ .

**Example 1.11.17.** 1. Every nilpotent group, in particular every abelian group, is solvable.

2. A subgroup of a solvable group is solvable.

3. A quotient of a solvable group is solvable.

**Lemma 1.11.18.** *If  $N \triangleleft G$  and  $G/N$  are solvable, then so is  $G$ .*

**Example 1.11.19.** Let  $|G| = pq$  for  $p < q$  primes. If  $H$  is the Sylow  $q$ -group, then  $G$  is solvable since  $H$  and  $G/H$  are cyclic, hence solvable.

For a group of this form,  $G$  is nilpotent if and only if  $G$  is abelian (in which case it is cyclic). Thus for example,  $S_3$  is solvable but not nilpotent.

## 1.12 Symmetric and alternating groups

Recall that  $S_n$  is the symmetric group on the indices  $\{1, 2, \dots, n\}$ . Elements of  $S_n$  are called *permutations*.

**Definition 1.12.1** (Cycle / transposition). An element  $\sigma \in S_n$  is a  $k$ -cycle if there exist  $k$  distinct indices  $a_1, \dots, a_k$  with  $\sigma(a_j) = a_{j+1}$  and every other element fixed by  $\sigma$ . (Here  $a_{k+1} = a_1$ .)

A *transposition* is a 2-cycle.

**Notation.** A cycle is written  $\sigma = (a_1, a_2, \dots, a_k)$ . This is the same as  $(a_2, \dots, a_k, a_1)$ , etc.

**Example 1.12.2.** The elements of  $S_3$  are

$$\text{id}, \quad (1, 2), \quad (1, 3), \quad (2, 3), \quad (1, 2, 3), \quad (1, 3, 2).$$

**Proposition 1.12.3.** Every  $\sigma \in S_n$  can be written as a product of disjoint cycles. Moreover, this is unique up to rearrangement and internal cycling of indices.

*Proof.* Let  $H$  be the cyclic subgroup generated by  $\sigma$ . The natural action of  $S_n$  on the set  $X = \{1, \dots, n\}$  restricts to an action of  $H$  on  $X$ . Let  $X_1, \dots, X_m$  be all  $H$  orbits in  $X$ . Then  $\sigma$  permutes cyclically the elements of every orbit  $X_i$ . Therefore,  $\sigma$  is the product of disjoint cycles  $\sigma_1 \cdots \sigma_m$ .  $\square$

**Definition 1.12.4** (Cycle type). Let  $\sigma \in S_n$  and write

$$\sigma = (a_{1,1}, \dots, a_{1,k_1})(a_{2,1}, \dots, a_{2,k_2}) \cdots (a_{r,1}, \dots, a_{r,k_r}),$$

with  $k_1 \geq k_2 \geq \dots \geq k_r$  and  $k_1 + \dots + k_r = n$ . The *cycle type* of  $\sigma$  is the  $r$ -tuple  $(k_1, \dots, k_r)$ .

**Notation.** It is convenient to drop 1's from the cycle type if the degree  $n$  of the symmetric group  $S_n$  is understood. For example,  $(12)(34) = (12)(34)(5) \in S_5$  has cycle type  $(2, 2, 1)$  or simply  $(2, 2)$ .

Let  $\sigma = (a_1, a_2, \dots, a_k) \in S_n$  be a  $k$ -cycle and  $\tau \in S_n$ . Then  $\tau\sigma\tau^{-1} = (b_1, b_2, \dots, b_k)$  is a  $k$ -cycle, where  $b_i = \tau(a_i)$  for all  $i$ .

**Proposition 1.12.5.** Two permutations  $\sigma, \tau \in S_n$  are conjugate if and only if they have the same cycle type.

*Proof.* If  $\sigma = \sigma_1 \cdots \sigma_m$  is the product of disjoint cycles, where  $\sigma_i$  is a  $k_i$ -cycle, then  $\tau\sigma\tau^{-1} = (\tau\sigma_1\tau^{-1}) \cdots (\tau\sigma_m\tau^{-1})$  is the product of disjoint  $k_i$ -cycles  $\tau\sigma_i\tau^{-1}$ , i.e.,  $\sigma$  and  $\tau\sigma\tau^{-1}$  have the same cycle type.

Conversely, it is sufficient to prove that the two  $k$ -cycles  $\sigma = (a_1, a_2, \dots, a_k)$  and  $\sigma' = (b_1, b_2, \dots, b_k)$  are conjugate in  $S_k$ . If  $\tau \in S_k$  is such that  $\tau(a_i) = b_i$  for all  $i$ , then  $\sigma' = \tau\sigma\tau^{-1}$ .  $\square$

**Example 1.12.6.** The  $k$ -cycle  $(a_1, \dots, a_k)$  is a product of  $k - 1$  transpositions

$$(a_1, a_k)(a_1, a_{k-1}) \cdots (a_1, a_2).$$

**Proposition 1.12.7.** Every element in  $S_n$  is a product of transpositions.

**Definition 1.12.8** (Permutation representation). The *(complex) permutation representation* of  $S_n$  is the homomorphism  $\pi : S_n \rightarrow GL_n(\mathbb{C})$  given by

$$\pi(\sigma) = (\delta_{\sigma(i),j}).$$

**Definition 1.12.9** (Sign homomorphism / alternating group). The *sign homomorphism* of  $S_n$  is  $\text{sgn} = \det \circ \pi : S_n \rightarrow \{\pm 1\}$ . The *alternating group* is  $A_n = \text{Ker sgn} \triangleleft S_n$ . We say that  $\sigma$  from  $S_n$  is *even* if  $\sigma \in A_n$  and *odd* otherwise.

If  $n > 1$ , we have  $|A_n| = n!/2$ , so there are  $n!/2$  even and  $n!/2$  odd permutations.

**Proposition 1.12.10.** *If  $\sigma$  can be written as a product of  $k$  transpositions, then  $\text{sgn } \sigma = (-1)^k$ .*

**Lemma 1.12.11.**  *$A_n$  is generated by 3-cycles.*

*Proof.* It suffices to write any product of two transpositions in terms of 3-cycles, as elements of  $A_n \leq S_n$  can be written as products of transpositions, and being in  $A_n$  requires these products to have an even number of factors. For this, we have

$$\begin{aligned} (a, b)(a, b) &= \text{id}, \\ (a, b)(b, c) &= (a, b, c), \\ (a, b)(c, d) &= (a, b, c)(b, c, d). \end{aligned} \quad \square$$

**Lemma 1.12.12.** *If  $n \geq 5$ , then any two 3-cycles in  $A_n$  are conjugate.*

*Proof.* Let  $\sigma, \tau \in A_n$  be 3-cycles. Then there exists  $\rho \in S_n$  with  $\tau = \rho\sigma\rho^{-1}$ . If  $\rho \in A_n$ , we are done. Otherwise, suppose  $\sigma = (a, b, c)$ . Since  $n \geq 5$ , there are indices  $d, e$  disjoint from  $\sigma$ . Then  $\rho' = \rho(d, e) \in A_n$ , and

$$\rho'\sigma\rho'^{-1} = \rho(d, e)(a, b, c)(d, e)\rho^{-1} = \rho(a, b, c)\rho^{-1} = \tau. \quad \square$$

**Definition 1.12.13** (Simple group). A group  $G \neq 1$  is *simple* if its only normal subgroups are 1 and  $G$ .

**Example 1.12.14.** If  $G$  is abelian or solvable, then  $G$  is simple if and only if  $G \cong C_p$ .

**Theorem 1.12.15.** *If  $n \geq 5$ , then  $A_n$  is simple.*

*Proof.* Let  $N \triangleleft A_n$  be non-trivial. It suffices to show that  $N$  contains a 3-cycle. Pick some  $\sigma \in N$ . If  $\sigma$  is not a 3-cycle, then since (single) transpositions are not in  $A_n$ ,  $\sigma$  moves at least four indices. Suppose  $\sigma$  contains a  $k$ -cycle for some  $k \geq 4$ , say  $\sigma = (12 \cdots k)\tau$  by relabeling (conjugation). Then

$$\sigma(123)\sigma^{-1}(123)^{-1} = (234)(132) = (142) \in N.$$

Suppose  $\sigma = (123)(456)\tau$  for some  $\tau$  which is a product of 3-cycles and transpositions. Then

$$\sigma(124)\sigma^{-1}(124)^{-1} = (235)(142) = (14352) \in N,$$

so we are done by the previous case.

Suppose  $\sigma = (123)\tau$  for some  $\tau$  which is a product of transpositions. Then  $\sigma\sigma = (132) \in N$ .

Finally, suppose  $\sigma = (12)(34)\tau$  for some  $\tau$  which is a product of transpositions. Then

$$\pi = \sigma(123)\sigma^{-1}(123)^{-1} = (214)(132) = (13)(24) \in N$$

and  $\pi(135)\pi^{-1}(135)^{-1} = (315)(153) = (135) \in N$ . □

**Corollary 1.12.16.**  $S_n$  is not solvable for  $n \geq 5$ .

**Remark 1.12.17.** The group  $A_3 \cong C_3$  is also simple, while groups  $A_1$  and  $A_2$  are trivial. By order,  $A_5$  is the smallest non-abelian simple group.

**Proposition 1.12.18.**  $S_n$  is solvable for  $n \leq 4$ .

*Proof.*  $\{1, (12)(34), (13)(24), (14)(23)\}$  is a normal subgroup of  $S_4$ . □

### 1.13 Semidirect products

**Definition 1.13.1** (Semidirect product). Let  $N$  and  $K$  be groups and  $f : K \rightarrow \text{Aut } N$  be a homomorphism. The (*external*) *semidirect product*  $N \rtimes_f K$  is the group on  $N \times K$  with operation

$$(h_1, k_1)(h_2, k_2) = (h_1 f(k_1)(h_2), k_1 k_2).$$

**Example 1.13.2.** 1. If  $f$  is the trivial homomorphism, then  $N \rtimes_f K = N \times K$ .

2. Let  $N = C_n$  and  $K = C_2$ . There is a non-trivial homomorphism  $f : C_2 \rightarrow \text{Aut } C_n$  which sends the non-identity element of  $C_2$  to the map  $g \mapsto g^{-1}$ . This gives us the group  $C_n \rtimes_f C_2$  with

$$(r_1, s_1)(r_2, s_2) = \begin{cases} (r_1 r_2, s_1 s_2), & \text{if } s_1 = e; \\ (r_1 r_2^{-1}, s_1 s_2), & \text{otherwise.} \end{cases}$$

This is the *dihedral group*, denoted  $D_{2n}$ . The semidirect product  $C_\infty \rtimes_f C_2$  is denoted  $D_\infty$ . It has order  $2n$ .

3. If  $p < q$  are primes and  $q \equiv 1 \pmod{p}$ , then there is a non-trivial  $f : C_p \rightarrow \text{Aut } C_q$ . Then  $C_q \rtimes_f C_p$  is a non-abelian group of order  $pq$ .

**Proposition 1.13.3.**  $N \triangleleft (N \rtimes_f K)$ .

**Definition 1.13.4** (Internal semidirect product). Let  $N \triangleleft G$  and  $K \subset G$  be subgroups. If  $N \cap K = 1$  and  $G = NK$ , then we say that  $G$  is the *internal semidirect product* of  $N$  and  $K$ .

**Remark 1.13.5.** If  $G$  is finite, then the last condition is equivalent to  $|G| = |N||K|$ .

**Proposition 1.13.6.** If  $G$  is the internal semidirect product of  $N$  and  $K$ , then  $G \cong N \rtimes_f K$  for some homomorphism  $f : K \rightarrow \text{Aut } N$ .

**Proposition 1.13.7.** If  $H$  is a group, then  $\text{Aut } H \subset \text{Inn } G$  for some group  $G$  containing  $H$ .

*Proof.* We can take  $G = H \rtimes_f \text{Aut } H$  with  $f : \text{Aut } H \rightarrow \text{Aut } H$  the identity. □

## 1.14 Groups of small order

In this section we classify groups of order  $n = |G| \leq 15$  up to isomorphism.

Throughout, write  $C_n = \mathbb{Z}/n\mathbb{Z}$  and assume  $p < q$  are primes.

**Proposition 1.14.1.** *The only group of order 1 is 1.*

**Proposition 1.14.2.** *If  $n = p$ , then  $G \cong C_p$ .*

Let  $p$  be a prime integer. A finite group  $G$  is called *elementary abelian  $p$ -group* if  $G$  is an abelian  $p$ -group such that  $a^p = 1$  for all  $a \in G$ . Such  $G$  can be viewed as a vector space over the field  $\mathbb{Z}/p\mathbb{Z}$ , hence it has a basis. It follows that  $G \cong C_p \times C_p \times \cdots \times C_p$ .

**Proposition 1.14.3.** *If  $n = p^2$  for  $p$  prime, then  $G \cong C_{p^2}$  or  $G \cong C_p \times C_p$ .*

*Proof.* If  $G$  contains an element of order  $p^2$ , then  $G \cong C_{p^2}$ .

Otherwise, every non-identity element of  $G$  has order  $p$ , i.e.,  $G$  is an elementary abelian  $p$ -group. It follows that  $G \cong C_p \times C_p$ .  $\square$

**Proposition 1.14.4.** *If  $n = pq$  and  $q \not\equiv 1 \pmod{p}$ , then  $G \cong C_{pq}$ .*

*Proof.* This is Proposition 1.10.9.  $\square$

**Proposition 1.14.5.** *If  $q \equiv 1 \pmod{p}$  instead, then  $G \cong C_{pq}$  or  $G \cong C_q \rtimes_f C_p$  for  $f$  non-trivial.*

*Proof.* The Sylow  $q$ -subgroup  $Q \cong C_q$  is normal in  $G$ , while the number of Sylow  $p$ -subgroups is either 1 or  $q$ . If  $n_p = 1$ , then  $G \cong C_q \times C_p \cong C_{pq}$  by Proposition 1.10.9.

If  $n_p = q$ , then fix some Sylow  $p$ -subgroup  $P$ . The subgroups  $Q$  and  $P$  meet the conditions for  $G$  to be an internal semidirect product, i.e.  $G = Q \rtimes_f P$  for some non-trivial  $f : P \rightarrow \text{Aut } Q$ .  $\square$

**Corollary 1.14.6.** *If  $p$  is an odd prime and  $|G| = 2p$ , then  $G \cong C_{2p}$  or  $G \cong D_{2p}$ .*

*Proof.* By Proposition 1.14.5, either  $G \cong C_{2p}$  or  $G \cong C_p \rtimes_f C_2$  for some non-trivial  $f : C_2 \rightarrow \text{Aut } C_p$ . The only such  $f$  is the one which sends the generator of  $C_2$  to the inversion automorphism of  $C_p$ , which gives the dihedral group  $D_{2p}$ .  $\square$

Using these results, we can classify groups of order up to  $n = 15$  except for  $n = 8$  and  $n = 12$ .

- |                          |                          |
|--------------------------|--------------------------|
| 1. 1                     | 9. $C_9, C_3 \times C_3$ |
| 2. $C_2$                 | 10. $C_{10}, D_{10}$     |
| 3. $C_3$                 | 11. $C_{11}$             |
| 4. $C_4, C_2 \times C_2$ | 13. $C_{13}$             |
| 5. $C_5$                 | 14. $C_{14}, D_{14}$     |
| 6. $C_6, D_6 \cong S_3$  | 15. $C_{15}$             |
| 7. $C_7$                 |                          |

**Proposition 1.14.7.** *The groups of order 8 are  $C_8$ ,  $C_4 \times C_2$ ,  $C_2 \times C_2 \times C_2$ ,  $D_8$ , and*

$$Q_8 = \left\{ \pm \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \pm \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \pm \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \pm \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \right\}.$$

*Proof.* If  $G$  has an element of order 8, then  $G \cong C_8$ .

If  $G$  has no elements of order 4 or 8, so every non-identity element has order 2, then repeated use of the direct product theorem shows that  $G \cong C_2 \times C_2 \times C_2$ .

If  $h \in G$  has order 4 and no element of  $G$  has order 8, then let  $H = \langle h \rangle \triangleleft G$  and pick  $k \in G \setminus H$ .

If  $k$  has order 2, then  $K = \langle k \rangle$  intersects  $H$  trivially and  $|G| = |H||K|$ , so  $G = H \rtimes_f K$  for some  $f : K \rightarrow \text{Aut } H$ . There are two possibilities for  $f$ , which correspond to  $C_4 \times C_2$  if  $f$  is trivial and  $D_8$  if  $f$  sends  $k$  to the inversion automorphism.

If no  $k \in G \setminus H$  has order 2, so then every  $k \in G \setminus H$  has order 4, we have  $k^2 = h^2$  for all  $k \in G \setminus H$  since  $(kH)(kH) = H$  in  $G/H$  and  $k^2$  has order 2. From this we can deduce that  $h^2 \in Z$  and  $hk = kh^3$ , which is enough to deduce the multiplication table of  $G$ . One can check that  $Q_8$  has the same multiplication table.  $\square$

**Proposition 1.14.8.** *The groups of order 12 are  $C_{12}$ ,  $C_6 \times C_2$ ,  $D_{12}$ ,  $A_4$ , and  $C_3 \rtimes_f C_4$  for*

$$C_3 = \{1, a, a^2\}; \quad C_4 = \{1, b, b^2, b^3\}; \quad f(b)(a) = a^{-1}.$$

*Proof.* Let  $P \subset G$  be a Sylow 3-subgroup.

If  $P$  is not normal, then there are 4 Sylow 3-subgroups, each of which contains 2 elements of order 3. Then there are only four elements of  $G$  not of order 3, so the unique Sylow 2-subgroup  $Q$  of 4 elements must contain all of them, and thus  $Q$  is normal. Hence  $G \cong Q \rtimes_f P$  for some  $f : P \rightarrow \text{Aut } Q$ .

If  $Q \cong C_4$ , then we seek homomorphisms  $f : C_3 \rightarrow \text{Aut } C_4$ . There are two elements of  $\text{Aut } C_4$ , and only the identity satisfies  $\sigma^3 = 1$ , so the only possible choice of  $f$  is the trivial map. This gives  $G \cong C_4 \times C_3 \cong C_{12}$ .

If  $Q \cong C_2 \times C_2$ , then  $\text{Aut } Q \cong S_3$ , which has three elements  $\sigma$  satisfying  $\sigma^3 = 1$ . Thus we have three possible choices of  $f : P \rightarrow \text{Aut } Q$ . If  $f$  is trivial, then  $G \cong C_2 \times C_2 \times C_3 \cong C_6 \times C_2$ . Otherwise, if  $P = \{1, a, a^2\}$  and  $Q = \{1, b, c, bc\}$ , then without loss of generality  $f(a)(b) = c$  and  $f(a)(c) = bc$  (otherwise, relabel elements). The multiplication table is then completely determined, and it matches that of  $A_4$ .

Now suppose  $P$  is normal and let  $Q$  be a Sylow 2-subgroup of order 4. In this case  $G \cong P \rtimes_g Q$  for some  $g : Q \rightarrow \text{Aut } P$ .

If  $Q \cong C_4$ , then  $\text{Aut } P$  has two elements, both satisfying  $\sigma^4 = 1$ , so there are two choices for  $g$ . If  $g$  is trivial, then we get  $G \cong C_3 \times C_4 \cong C_{12}$  again. Otherwise, if  $P = \{1, a, a^2\}$  and  $Q = \{1, b, b^2, b^3\}$ , then elements of  $G$  have the form  $a^i b^j$ , and multiplication is determined by  $ba = a^{-1}b$ . This is the last group of the list.

If  $Q \cong C_2 \times C_2$ , then there are four homomorphisms  $g : Q \rightarrow \text{Aut } P$ , but three of them are the same up to an isomorphism of  $Q$  (i.e. by relabeling generators). The trivial  $g$  produces  $G \cong C_6 \times C_2$ , while the non-trivial choices of  $g$  turn out to produce  $D_{12}$ .  $\square$

**Remark 1.14.9.** There are 14 isomorphism classes of groups of order 16. There are 2,328 isomorphism classes of groups of order  $2^7 = 128$ . There are 49,487,367,289 isomorphism classes of groups of order  $2^{10} = 1024$ .

## 1.15 Exact sequences

**Definition 1.15.1** (Exact sequence). A sequence of group homomorphisms

$$G_0 \xrightarrow{f_1} G_1 \xrightarrow{f_2} \cdots \xrightarrow{f_n} G_n$$

is *exact* if  $\text{Im } f_i = \text{Ker } f_{i+1}$  for all  $i = 1, \dots, n-1$ .

**Proposition 1.15.2.** 1.  $f : G \rightarrow H$  is injective if and only if  $1 \rightarrow G \xrightarrow{f} H$  is exact.

2.  $f : G \rightarrow H$  is surjective if and only if  $G \xrightarrow{f} H \rightarrow 1$  is exact.

**Definition 1.15.3** (Short exact sequence). A *short exact sequence* is an exact sequence

$$1 \longrightarrow H \xrightarrow{\alpha} G \xrightarrow{\beta} F \longrightarrow 1,$$

i.e.  $\alpha$  is injective,  $\beta$  is surjective, and  $\text{Im } \alpha = \text{Ker } \beta$ . Then  $H$  identifies with  $\text{Im } \alpha \triangleleft G$  and  $F \cong G/H$ .

**Proposition 1.15.4.** If  $H \triangleleft G$ , then the sequence

$$1 \longrightarrow H \xhookrightarrow{i} G \xrightarrow{\pi} G/H \longrightarrow 1$$

is exact.

**Definition 1.15.5** (Split exact sequence). A short exact sequence

$$1 \longrightarrow H \xrightarrow{\alpha} G \xleftarrow[\gamma]{\beta} F \longrightarrow 1$$

is *split* (or *right split*) if there exists a homomorphism  $\gamma : F \rightarrow G$  such that  $\beta \circ \gamma = \text{id}_F$ .

**Theorem 1.15.6.** The short exact sequence

$$\begin{array}{ccccccc} 1 & \longrightarrow & H & \xrightarrow{\alpha} & G & \xleftarrow[\gamma]{\beta} & F \longrightarrow 1 \\ & & & & \uparrow i & \nearrow \beta_K & \\ & & & & K & & \end{array}$$

is split if and only if there is a subgroup  $K \subset G$  such that  $\beta|_K : K \rightarrow F$  is an isomorphism. In this case,  $G \cong H \rtimes_{\varphi} F$  with  $\varphi : F \rightarrow \text{Aut } H$  given by  $\varphi(f)(h) = hg^{-1}$ , where  $g = \gamma(f)$ .

*Proof.* ( $\implies$ ) Let  $K = \text{Im } \gamma$ . If  $f \in F$  and  $k = \gamma(f) \in K$ , then  $\beta|_K(k) = \beta(\gamma(f)) = f$ , so  $\beta|_K : K \rightarrow F$  is surjective. If  $k \in \text{Ker } \beta|_K$ , then  $k = \gamma(f)$  for some  $f \in F$  since  $K = \text{Im } \gamma$ , and then  $f = \beta(\gamma(f)) = \beta(k) = 1$ . This means  $k = \gamma(1) = 1$ , so  $\text{Ker } \beta|_K$  must be trivial.

( $\Leftarrow$ ) Suppose such a  $K$  exists, so  $\beta|_K : K \rightarrow F$  is an isomorphism. Take  $\gamma = \beta|_K^{-1} \circ i$ .

If these conditions are met, then regard  $K \cong \gamma(K)$  as a subgroup of  $G$ . We have that  $H = \text{Ker } \beta \triangleleft G$ , and  $H \cap K = 1$  since  $\beta(H) = 1$  and  $\beta|_K : K \rightarrow F$  is an isomorphism. Finally, if  $g \in G$  and  $f = \beta(g) \in F$ , then  $k = \gamma(f) \in K$  and  $h = gk^{-1} \in H$  satisfy  $hk = g$ , so  $G = HK$ . Hence  $G$  is the internal semidirect product of  $H$  and  $K$ , so  $G \cong H \rtimes_{\varphi} F$  for some  $\varphi : F \rightarrow \text{Aut } H$ . To determine  $\varphi$ , let  $g = \gamma(f) \in K$  and  $h \in H$ . Then  $gh = \varphi(g)(h) \cdot g$ , so  $\varphi(g)(h) = ghg^{-1}$ , as required.  $\square$

**Example 1.15.7.** Let  $G$  be a non-abelian group of order 8, and let  $h \in G$  be an element of order 4. Then  $H = \langle h \rangle \triangleleft G$ , so we have a short exact sequence

$$1 \longrightarrow H \hookrightarrow G \twoheadrightarrow G/H \longrightarrow 1.$$

If there is an element of order 2 in  $G \setminus H$ , then the sequence splits and we obtain  $G \cong D_8$  via the theorem. Otherwise, there is no splitting, and we obtain  $Q_8$  as before.

## 1.16 Free groups

**Definition 1.16.1** (Words). Let  $X$  be a set, called the *alphabet*. The elements of  $X$  are referred to as *letters*. Form an inverse alphabet  $\bar{X}$  of formal symbols  $\{\bar{x} \mid x \in X\}$ .

For  $n \geq 0$ , a *word of length  $n$  on  $X$*  is a sequence of  $n$  letters (not necessarily distinct) from  $X \cup \bar{X}$ .

Concatenation of words  $\mathbf{w}$  and  $\mathbf{v}$  to get a new word  $\mathbf{wv}$  defines an associative non-commutative binary operation on the set of words on a given alphabet. The empty word  $\epsilon$  is the identity for this operation, but no element (other than  $\epsilon$ ) has an inverse.

**Definition 1.16.2** (Truncation / irreducible word). Let  $\mathbf{w}$  be a word on  $X$  of the form  $\mathbf{u}a\bar{\mathbf{a}}\mathbf{v}$  or  $\mathbf{u}\bar{a}\mathbf{a}\mathbf{v}$ , where  $\mathbf{u}, \mathbf{v}$  are words on  $X$  and  $a \in X$  is a letter. The word  $\mathbf{w}' = \mathbf{uv}$  is a *truncation* of  $\mathbf{w}$ .

If  $\mathbf{w}$  is a word on  $X$ , then  $\mathbf{w}$  is *irreducible* if there is no word  $\mathbf{u}$  on  $X$  which is a truncation of  $\mathbf{w}$ .

For two words  $\mathbf{u}, \mathbf{v}$  on  $X$ , write  $\mathbf{u} \sim \mathbf{v}$  if there is a sequence of words

$$\mathbf{u} = \mathbf{w}_0, \mathbf{w}_1, \dots, \mathbf{w}_n = \mathbf{v}$$

such that for each  $i$ , one of the words  $\mathbf{w}_i, \mathbf{w}_{i+1}$  is a truncation of the other. Then  $\sim$  is an equivalence relation on the set of words on  $X$ .

**Proposition 1.16.3.** If  $\mathbf{u}_1 \sim \mathbf{v}_1$  and  $\mathbf{u}_2 \sim \mathbf{v}_2$ , then  $\mathbf{u}_1\mathbf{u}_2 \sim \mathbf{v}_1\mathbf{v}_2$ .

**Theorem 1.16.4.** Each equivalence class of words on  $X$  contains exactly one irreducible word.

*Proof.* For existence, let  $\mathbf{w}$  be a word of minimum length in a given equivalence class. Since truncation reduces the length of a word by 2,  $\mathbf{w}$  must be irreducible.

For uniqueness, let  $\mathbf{u}$  and  $\mathbf{v}$  be irreducible words in the same equivalence class, and write down a sequence  $\mathbf{u} = \mathbf{w}_0, \mathbf{w}_1, \dots, \mathbf{w}_n = \mathbf{v}$  as above. To show that  $\mathbf{u} = \mathbf{v}$ , we induct on  $n$  and then the total length of the words  $\mathbf{w}_0, \dots, \mathbf{w}_n$ . When  $n = 0$ , we have  $\mathbf{u} = \mathbf{w}_0 = \mathbf{v}$ . Now consider  $n \geq 1$  and look at a longest word  $\mathbf{w}_k$  in the sequence. Then  $\mathbf{w}_{k-1}$  and  $\mathbf{w}_{k+1}$  are necessarily truncations of  $\mathbf{w}_k$ .

If the truncations are

$$st\bar{b}b\mathbf{u} \leftarrow sa\bar{a}t\bar{b}b\mathbf{u} \rightarrow sa\bar{a}t\mathbf{u},$$

then we replace  $\mathbf{w}_k$  with  $\mathbf{stu}$  and have truncations  $\mathbf{w}_{k-1} \rightarrow \mathbf{w}_k$  and  $\mathbf{w}_{k+1} \rightarrow \mathbf{w}_k$  instead. This does not change the number of words  $n$  in the sequence, but it does reduce the total length of all words in the sequence, so we can apply the inductive hypothesis.

If the truncations are

$$\mathbf{sat} \leftarrow \mathbf{sa\bar{a}at} \rightarrow \mathbf{sat} \quad \text{or} \quad \mathbf{st} \leftarrow \mathbf{aa\bar{a}t} \rightarrow \mathbf{st},$$

then we can reduce the number of words  $n$  in the sequence by omitting  $\mathbf{w}_k$  and  $\mathbf{w}_{k+1}$ .

We can obtain every other case by swapping the roles of letters and inverse letters.  $\square$

**Definition 1.16.5** (Free group). Let  $X$  be an alphabet. The *free group on  $X$* , denoted  $\text{Free}(X)$ , is the set of all equivalence classes of words on  $X$  with the concatenation operation.

**Example 1.16.6.** 1. If  $X = \emptyset$ , then  $\text{Free}(X) = 1$ .

2. If  $X = \{a\}$ , then  $\text{Free}(X) \cong \mathbb{Z}$  is the cyclic group generated by  $[a]$ .

3. If  $|X| \geq 2$  with  $a, b \in X$  distinct, then  $ab$  and  $ba$  are distinct irreducible words. Hence  $[ab] \neq [ba]$ , so  $\text{Free}(X)$  is non-abelian.

**Notation.** In  $\text{Free}(X)$ , we write  $a^{-1}$  for  $\bar{a}$  when  $a \in X$ .

For convenience, we will write  $\mathbf{w}$  for  $[\mathbf{w}]$  and work on words themselves whenever possible.

**Theorem 1.16.7** (Universal property of free groups). *Let  $X$  be a set,  $G$  be a group, and  $f : X \rightarrow G$  be a set function. Then there is a unique group homomorphism  $\bar{f} : \text{Free}(X) \rightarrow G$  such that  $\bar{f}(x) = f(x)$  for all  $x \in X$ .*

*Proof.* Let  $\mathbf{w} \in \text{Free}(X)$  and write  $\mathbf{w} = b_1 \cdots b_n$  where  $b_i = x_i^{\epsilon_i}$  for some  $x_i \in X$  and  $\epsilon_i \in \{\pm 1\}$ . Then

$$\bar{f}(\mathbf{w}) = \bar{f}(b_1) \cdots \bar{f}(b_n) = f(x_1)^{\epsilon_1} \cdots f(x_n)^{\epsilon_n},$$

which shows that if  $\bar{f}$  exists, then it is unique and must be given by this formula.

To show existence, we can define  $\bar{f}$  using this formula, provided it is well-defined on  $\text{Free}(X)$ . For this, note that  $\bar{f}$  is unchanged by truncations, hence whenever two words are equivalent.  $\square$

**Corollary 1.16.8.** *Let  $X \subset G$  generate  $G$ . Then  $G \cong \text{Free}(X)/N$  for some  $N \triangleleft \text{Free}(X)$ .*

*Proof.* The inclusion function  $i : X \hookrightarrow G$  extends to a homomorphism  $f : \text{Free}(X) \rightarrow G$ . That  $X$  generates  $G$  means that  $f$  is surjective, so  $\text{Free}(X)/\text{Ker } f \cong G$ .  $\square$

**Notation.** If  $G$  is a group and  $S \subset G$  is a subset, then

$$\langle\langle S \rangle\rangle = \left\langle \bigcup_{g \in G} gSg^{-1} \right\rangle$$

is the smallest normal subgroup of  $G$  containing  $S$ .

**Definition 1.16.9** (Presentation / finite presentation). Let  $G$  be a group and  $X \subset G$  generate  $G$ . Choose a subset  $R \subset \text{Free}(X)$  such that  $\langle\langle R \rangle\rangle = \text{Ker}(\text{Free}(X) \rightarrow G)$ . Then we write

$$G \cong \langle X \mid R \rangle = \text{Free}(X) / \langle\langle R \rangle\rangle.$$

This is a *presentation* for  $G$ . We say that the presentation is *finite* if  $X$  and  $R$  are finite.

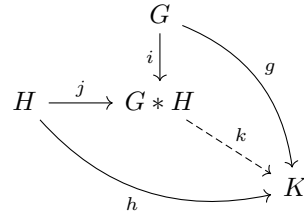
**Example 1.16.10.** 1.  $\langle \sigma \mid \sigma^n \rangle \cong \mathbb{Z}/n\mathbb{Z}$

2.  $\langle \sigma, \tau \mid \sigma^n, \tau^2, \tau\sigma\tau\sigma \rangle \cong D_{2n}$

Let  $G$  and  $H$  be two groups. Fix presentations  $G = \langle X \mid R \rangle$  and  $H = \langle Y \mid S \rangle$  and set

$$G * H = \langle X \cup Y \mid R \cup S \rangle.$$

**Theorem 1.16.11** (Universal property of free products). *Let  $i, j : G, H \rightarrow G * H$  be the natural maps and let  $g, h : G, H \rightarrow K$  be homomorphisms to some group  $K$ . Then there is a unique homomorphism  $k : G * H \rightarrow K$  such that the following diagram commutes.*



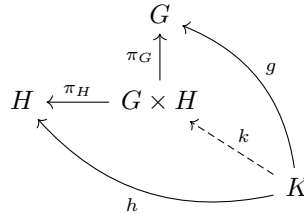
It follows that  $G * H$  does not depend on the presentations. This group is called *free product (coproduct)* of  $G$  and  $H$ .

**Example 1.16.12.** 1.  $G * 1 \cong G$  and  $1 * H \cong H$ .

2.  $C_2 * C_2 \simeq D_\infty$ .

The universal property of free products is closely related to the following result for direct products.

**Theorem 1.16.13** (Universal property of direct products). *Let  $\pi_G, \pi_H : G \times H \rightarrow G, H$  be the projections and let  $g, h : K \rightarrow G, H$  be homomorphisms from some group  $K$ . Then there is a unique homomorphism  $k : K \rightarrow G \times H$  such that the following diagram commutes.*





## 2 Categories and Functors

### 2.1 Definitions and basic properties

**Definition 2.1.1** (Category). A *category*  $\mathcal{C}$  consists of a collection (more formally a class)  $\text{Ob } \mathcal{C}$  of *objects*, a collection  $\text{Mor } \mathcal{C}$  of *morphisms* (arrows) between objects. Every morphism  $f$  has the *source* object  $X = s(f)$  and the *target* object  $Y = t(f)$ . We write  $f : X \rightarrow Y$ . Moreover, there is a composition operation, which forms from morphisms  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  a morphism  $g \circ f : X \rightarrow Z$ , such that

- (i) if  $W \xrightarrow{f} X \xrightarrow{g} Y \xrightarrow{h} Z$  are morphisms, then  $h \circ (g \circ f) = (h \circ g) \circ f$ ;
- (ii) for any object  $X \in \text{Ob } \mathcal{C}$ , there is a (unique) identity morphism  $1_X : X \rightarrow X$  such that for any morphisms  $f : X \rightarrow Y$  and  $g : W \rightarrow X$ , we have

$$f = f \circ 1_X, \quad g = 1_X \circ g.$$

The collection of morphisms  $X \rightarrow Y$  is denoted  $\text{Mor}_{\mathcal{C}}(X, Y)$  or simply  $\text{Mor}(X, Y)$ .

**Definition 2.1.2** (Small / locally small category). A *small category* is a category  $\mathcal{C}$  for which  $\text{Ob } \mathcal{C}$  and  $\text{Mor } \mathcal{C}$  are sets. A *locally small category* is one for which we can only say that  $\text{Mor}_{\mathcal{C}}(X, Y)$  is a set for each pair of objects  $X, Y \in \mathcal{C}$ .

In what follows we will only consider locally small categories.

**Example 2.1.3.** 1. In **Set**, the category of sets, the morphisms are maps (functions).

2. In **Grp**, the category of groups, the morphisms are group homomorphisms.

3. Given a group  $G$ , we can form a category  $\underline{G}$  with  $\text{Ob } \underline{G} = \{*\}$  and  $\text{Mor}(*, *) = G$ . The composition in  $\underline{G}$  is the product in  $G$ .

4. Given a poset  $X$ , we can form a category  $\mathcal{C}$  with  $\text{Ob } \mathcal{C} = X$  and

$$\text{Mor}(x, x') = \begin{cases} \{(x, x')\} & x \geq x', \\ \emptyset & \text{otherwise.} \end{cases}$$

5. Given categories  $\mathcal{C}$  and  $\mathcal{D}$ , the *product category*  $\mathcal{C} \times \mathcal{D}$  has  $\text{Ob}(\mathcal{C} \times \mathcal{D}) = \text{Ob } \mathcal{C} \times \text{Ob } \mathcal{D}$  and  $\text{Mor}_{\mathcal{C} \times \mathcal{D}}((A, X); (B, Y)) = \text{Mor}_{\mathcal{C}}(A, B) \times \text{Mor}_{\mathcal{D}}(X, Y)$  in the natural way.

6. Given a category  $\mathcal{C}$ , the *dual category* (*opposite category*), denoted  $\mathcal{C}^{\text{op}}$  is the category with  $\text{Ob } \mathcal{C}^{\text{op}} = \text{Ob } \mathcal{C}$  and  $\text{Mor}_{\mathcal{C}^{\text{op}}}(X, Y) = \text{Mor}_{\mathcal{C}}(Y, X)$ . For disambiguation, we may write  $X^{\circ}$  to denote the copy of  $X$  in  $\mathcal{C}^{\text{op}}$ .

7. Given a category  $\mathcal{C}$ , the *arrow category*  $\text{Arr } \mathcal{C}$  has  $\text{Ob } \mathcal{C} = \text{Mor } \mathcal{C}$ . A morphism between  $f : X \rightarrow Y$  and  $f' : X' \rightarrow Y'$  is given by morphisms  $g : X \rightarrow X'$  and  $h : Y \rightarrow Y'$  such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow g & & \downarrow h \\ X' & \xrightarrow{f'} & Y' \end{array}$$

**Definition 2.1.4** (Isomorphism). A morphism  $f : X \rightarrow Y$  is an *isomorphism* if there exists a morphism  $g : Y \rightarrow X$  such that  $f \circ g = 1_Y$  and  $g \circ f = 1_X$ .

**Proposition 2.1.5.** *If  $f : X \rightarrow Y$  is an isomorphism with  $g : Y \rightarrow X$  as above, then  $g$  is unique and  $g$  is an isomorphism.*

**Notation.** If  $f$  is an isomorphism, then write  $f^{-1}$  for the morphism  $g$  above.

**Definition 2.1.6** (Subcategory / full subcategory). Let  $\mathcal{C}$  be a category. A category  $\mathcal{C}'$  is a *subcategory* of  $\mathcal{C}$  if  $\text{Ob } \mathcal{C}' \subset \text{Ob } \mathcal{C}$ ,  $\text{Mor}_{\mathcal{C}'}(X, Y) \subset \text{Mor}_{\mathcal{C}}(X, Y)$ , and the composition law in  $\mathcal{C}'$  is inherited from  $\mathcal{C}$ .

We say that  $\mathcal{C}'$  is a *full subcategory* of  $\mathcal{C}$  if  $\text{Mor}_{\mathcal{C}'}(X, Y) = \text{Mor}_{\mathcal{C}}(X, Y)$  for all  $X, Y \in \mathcal{C}'$ .

**Example 2.1.7.** 1. In **Grp**, the subcategory **Ab** of abelian groups is a full subcategory.

2. For any subclass  $A \subset \text{Ob } \mathcal{C}$ , there is a unique full subcategory  $\mathcal{C}'$  of  $\mathcal{C}$  such that  $\text{Ob } \mathcal{C}' = A$ .

**Definition 2.1.8** (Initial and terminal objects). Let  $X \in \mathcal{C}$  be an object.

1.  $X$  is *initial* if for every  $Y \in \mathcal{C}$ , there is a unique morphism  $X \rightarrow Y$ .
2.  $X$  is *terminal* (*final*) if for every  $W \in \mathcal{C}$ , there is a unique morphism  $W \rightarrow X$ .

**Proposition 2.1.9.** *If  $X \in \mathcal{C}$  is an object, then  $X$  is initial (terminal) in  $\mathcal{C}$  if and only if  $X$  is terminal (initial) in  $\mathcal{C}^{\text{op}}$ .*

**Example 2.1.10.** 1. In **Set**, the initial object is  $\emptyset$  and the terminal objects are singleton sets.

2. In **Grp**, the trivial group is initial and terminal.

3. Let  $G$  be a group. Then  $\underline{G}$  has no initial or terminal objects (unless  $G$  is trivial).

4. Let  $X$  be a poset and form  $\mathcal{C}$  on  $X$  as before. The initial object of  $\mathcal{C}$  is the maximum of  $X$  (if it exists), while the terminal object is the minimum of  $X$  (if it exists).

**Theorem 2.1.11.** *If  $X$  and  $X'$  are initial (terminal), then there is a unique isomorphism  $X \rightarrow X'$ , i.e.  $X$  and  $X'$  are canonically isomorphic.*

*Proof.* Let  $X$  and  $X'$  be initial, and let  $f : X \rightarrow X'$  and  $g : X' \rightarrow X$  be the unique morphisms. Then  $g \circ f : X \rightarrow X$  is a morphism  $X \rightarrow X$ , but since  $X$  is initial and  $\text{id}_X : X \rightarrow X$  is a morphism,  $g \circ f = 1_X$ . Similarly,  $f \circ g = 1_{X'}$ , so  $f$  and  $g$  are inverses and  $f$  is an isomorphism.  $\square$

## 2.2 Products and coproducts

Universal properties are applications of Theorem 2.1.11. As an example, we consider products.

**Definition 2.2.1** (Product of two objects). Let  $X, Y \in \mathcal{C}$ . An object  $X \times Y$  together with morphisms (projections)  $p : X \times Y \rightarrow X$  and  $q : X \times Y \rightarrow Y$  is a *product* of  $X$  and  $Y$  if for any

morphisms  $f : Z \rightarrow X$  and  $g : Z \rightarrow Y$ , there is a unique morphism  $h : Z \rightarrow X \times Y$  such that the following diagram commutes.

$$\begin{array}{ccc}
 & X & \\
 & \uparrow p & \\
 Y & \xleftarrow{q} & X \times Y \\
 & \nwarrow h & \nearrow f \\
 & Z & \\
 & \nwarrow g & \\
 & & 
 \end{array}$$

i.e.,  $p \circ h = f$  and  $q \circ h = g$ .

**Theorem 2.2.2.** Let  $X \times Y$  and  $\widetilde{X \times Y}$  be two products of  $X$  and  $Y$ , with projections  $p, q$  and  $\tilde{p}, \tilde{q}$ , respectively. Then there is a unique isomorphism  $h : X \times Y \rightarrow \widetilde{X \times Y}$  such that the following diagram commutes.

$$\begin{array}{ccc}
 X \times Y & \xrightarrow{p} & X \\
 q \downarrow & \searrow h & \uparrow \tilde{p} \\
 Y & \xleftarrow{\tilde{q}} & \widetilde{X \times Y}
 \end{array}$$

*Proof.* Fix  $X$  and  $Y$ , and consider a new category  $\mathcal{D}$  whose objects are diagrams of the form

$$\begin{array}{ccc}
 Z & \longrightarrow & X \\
 \downarrow & & \\
 Y & & 
 \end{array}$$

A morphism between diagrams for  $Z$  and  $Z'$  is given by a morphism  $Z \rightarrow Z'$  in  $\mathcal{C}$  such that the following diagram commutes.

$$\begin{array}{ccc}
 Z & \longrightarrow & X \\
 \downarrow & \searrow & \uparrow \\
 Y & \longleftarrow & Z'
 \end{array}$$

The diagram for  $Z = X \times Y$  is a terminal object in  $\mathcal{D}$ , which is unique up to unique isomorphism.  $\square$

The definition of the product can be restated as follows.

**Proposition 2.2.3.** Let  $p : X \times Y \rightarrow X$  and  $q : X \times Y \rightarrow Y$  be the projections. The function

$$\begin{aligned}
 \text{Mor}(Z, X \times Y) &\longrightarrow \text{Mor}(Z, X) \times \text{Mor}(Z, Y) \\
 h &\longmapsto (p \circ h, q \circ h)
 \end{aligned}$$

is a bijection.

**Definition 2.2.4** (Arbitrary product). Let  $\{X_i\}_{i \in I}$  be a family of objects. The *product* is the object  $\prod_i X_i$  along with morphisms  $p_j : \prod_i X_i \rightarrow X_j$  for which  $\text{Mor}(Z, \prod_i X_i) \cong \prod_i \text{Mor}(Z, X_i)$  with bijection  $h \mapsto \prod_i \{p_i \circ h\}$ .

**Definition 2.2.5** (Product morphism). Let  $f : X \rightarrow X'$  and  $g : Y \rightarrow Y'$  be morphisms. The *product morphism*  $f \times g : X \times Y \rightarrow X' \times Y'$  is given by the following commuting diagram.

$$\begin{array}{ccccc}
 X \times Y & \xrightarrow{q} & Y & & \\
 p \downarrow & \searrow f \times g & \searrow g & & \\
 X & & & & Y' \\
 & \searrow f & & \nearrow q' & \\
 & & X' & \xleftarrow{p'} & X' \times Y'
 \end{array}$$

**Example 2.2.6.** 1. In **Set**, the product is the ordinary Cartesian product.

2. In **Grp**, the product is the (external) direct product.

3. In **Ab**, the product is the (external) direct product. Within the context of abelian groups, it is also known as the *direct sum*, written  $G \oplus H$ .

4. In the category  $n \rightarrow n-1 \rightarrow \cdots \rightarrow 2 \rightarrow 1$ , we have  $i \times j = \max(i, j)$ .

**Definition 2.2.7** (Coproduct). The *coproduct* of  $X, Y \in \mathcal{C}$  is  $(X^\circ \times Y^\circ)^\circ$ , i.e. the product in  $\mathcal{C}^{\text{op}}$ . More explicitly, the coproduct is an object  $X * Y$  together with morphisms  $i : X \rightarrow X * Y$ ,  $j : Y \rightarrow X * Y$  such that given morphisms  $f : X \rightarrow Z$  and  $g : Y \rightarrow Z$ , there is a unique morphism  $h : X * Y \rightarrow Z$  such that  $h \circ i = f$  and  $h \circ j = g$ .

$$\begin{array}{ccccc}
 & X & & & \\
 & \downarrow i & & \searrow f & \\
 Y & \xrightarrow{j} & X * Y & \xrightarrow{h} & Z \\
 & \searrow g & & \nearrow &
 \end{array}$$

**Proposition 2.2.8.** *The function*

$$\begin{aligned}
 \text{Mor}(X * Y, Z) &\longrightarrow \text{Mor}(X, Z) \times \text{Mor}(Y, Z) \\
 h &\longmapsto (h \circ i, h \circ j)
 \end{aligned}$$

*is a bijection.*

**Definition 2.2.9** (Arbitrary coproduct). Let  $\{X_j\}_{j \in J}$  be a family of objects. The *coproduct* is the object  $\bigsqcup_j X_j$  along with morphisms  $i_k : X_k \rightarrow \bigsqcup_j X_j$  for which  $\text{Mor}(\bigsqcup_j X_j, Z) \cong \prod_j \text{Mor}(X_j, Z)$  with bijection  $h \mapsto \prod_j \{h \circ i_j\}$ .

**Example 2.2.10.** 1. In **Set**, the coproduct is the disjoint union  $X \sqcup Y$ .

2. In **Grp**, the coproduct is the free product  $G * H$ .

3. In **Ab**, the coproduct is the direct sum  $G \oplus H$ .

4. In  $n \rightarrow n-1 \rightarrow \cdots \rightarrow 2 \rightarrow 1$ , we have  $i * j = \min(i, j)$ .

**Definition 2.2.11** (Group object). Let  $\mathcal{C}$  be a category. A *group object* in  $\mathcal{C}$  is a quadruple  $(G, m, e, i)$  such that

- (i)  $G$  is an object;
- (ii)  $m : G \times G \rightarrow G$  is a morphism (corresponding to multiplication);
- (iii)  $e : F \rightarrow G$  is a morphism (corresponding to the identity element), where  $F$  is terminal;
- (iv)  $i : G \rightarrow G$  is a morphism (corresponding to inverses);
- (v) (associativity) the following diagram commutes;

$$\begin{array}{ccc} G \times G \times G & \xrightarrow{\text{id}_G \times m} & G \times G \\ \downarrow m \times \text{id}_G & & \downarrow m \\ G \times G & \xrightarrow{m} & G \end{array}$$

- (vi) (identity) the following diagrams commute, where  $\pi_G$  is projection onto  $G$ ;

$$\begin{array}{ccc} G \times F & \xrightarrow{\text{id}_G \times e} & G \times G \\ & \searrow \pi_G & \downarrow m \\ & & G \end{array} \quad \begin{array}{ccc} F \times G & \xrightarrow{e \times \text{id}_G} & G \times G \\ & \searrow \pi_G & \downarrow m \\ & & G \end{array}$$

- (vii) (inverse) the following diagrams commute.

$$\begin{array}{ccc} G & \xrightarrow{(\text{id}_G, i)} & G \times G \\ \downarrow & & \downarrow m \\ F & \xrightarrow{e} & G \end{array} \quad \begin{array}{ccc} G & \xrightarrow{(i, \text{id}_G)} & G \times G \\ \downarrow & & \downarrow m \\ F & \xrightarrow{e} & G \end{array}$$

**Example 2.2.12.** 1. The group objects in **Set** are the usual groups.

2. Group objects in **Top**, the category of topological spaces, are *topological groups*.

3. Group objects in **Grp** are abelian groups.

## 2.3 Functors

**Definition 2.3.1** (Functor). Let  $\mathcal{C}$  and  $\mathcal{D}$  be categories. A (*covariant*) *functor*  $F : \mathcal{C} \rightarrow \mathcal{D}$  is a collection of functions  $\text{Ob } \mathcal{C} \rightarrow \text{Ob } \mathcal{D}$  and  $\text{Mor}_{\mathcal{C}}(X, Y) \rightarrow \text{Mor}_{\mathcal{D}}(F(X), F(Y))$  such that

- (i)  $F(1_X) = 1_{F(X)}$  for  $X \in \mathcal{C}$ ;
- (ii) for morphisms  $X \xrightarrow{f} Y \xrightarrow{g} Z$  in  $\mathcal{C}$ , we have  $F(g \circ f) = F(g) \circ F(f)$ .

If instead  $F$  maps  $\text{Mor}_{\mathcal{C}}(X, Y)$  to  $\text{Mor}_{\mathcal{D}}(F(Y), F(X))$  and  $F(g \circ f) = F(f) \circ F(g)$  for all  $f, g$ , we say that  $F$  is a *contravariant functor*.

**Remark 2.3.2.** Contravariant functors are covariant functors  $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$  or  $\mathcal{C} \rightarrow \mathcal{D}^{\text{op}}$ , so it is only necessary to consider covariant functors.

**Lemma 2.3.3.** *If  $f : X \rightarrow Y$  is an isomorphism in  $\mathcal{C}$ , then  $F(f) : F(X) \rightarrow F(Y)$  is an isomorphism in  $\mathcal{D}$  and  $F(f)^{-1} = F(f^{-1})$ .*

**Corollary 2.3.4.** *If  $X \cong Y$  in  $\mathcal{C}$ , then  $F(X) \cong F(Y)$  in  $\mathcal{D}$ .*

**Example 2.3.5.** 1. The identity functor  $\text{id} : \mathcal{C} \rightarrow \mathcal{C}$  is defined by  $\text{id}(X) = X$  and  $\text{id}(f) = f$ .

2. Let  $Y \in \mathcal{D}$ . The constant functor  $c_Y : \mathcal{C} \rightarrow \mathcal{D}$  is given by  $c_Y(X) = Y$  and  $c_Y(f) = \text{id}_Y$ .

3. The *forgetful functor*  $\text{Forget} : \mathbf{Grp} \rightarrow \mathbf{Set}$  has  $\text{Forget}(G) = G$  and  $\text{Forget}(f) = f$ .

4. If  $\mathcal{C}' \subset \mathcal{C}$  is a subcategory, there is an *inclusion functor*  $I : \mathcal{C}' \hookrightarrow \mathcal{C}$ .

5. Let  $F : \mathbf{Grp} \rightarrow \mathbf{Ab}$  send a group  $G$  to its abelianization  $G/G'$ . Given  $f : G \rightarrow H$ , composing with the projection  $H \rightarrow H/H'$  gives a homomorphism  $\tilde{f} : G \rightarrow H/H'$ . Then  $\tilde{f}$  descends to a homomorphism  $G/G' \rightarrow H/H'$ , which we call  $F(f)$ . One can check that with this definition of  $F$  on morphisms,  $F$  is a functor.

6. Let  $G$  be a group. Then to give a functor  $\underline{G} \rightarrow \mathcal{C}$  is the same as to give an object  $X$  in  $\mathcal{C}$  together with a " $G$ -action" on  $X$ .

7. Let  $\mathcal{I}$  be a small category. A functor  $\mathcal{I} \rightarrow \mathcal{C}$  is a commutative diagram in  $\mathcal{C}$  of shape  $\mathcal{I}$ . For example,  $\text{Arr}(\mathcal{C})$  is the category of diagrams of shape  $\bullet \rightarrow \bullet$  in  $\mathcal{C}$ .

**Definition 2.3.6** (Represented / corepresented functors). Let  $\mathcal{C}$  be a locally small category and fix  $X \in \mathcal{C}$ . The *functor*  $\mathcal{C} \rightarrow \mathbf{Set}$  *represented by*  $X$  is

$$\begin{aligned} R^X(Y) &= \text{Mor}_{\mathcal{C}}(X, Y), \\ R^X(f)(g) &= f \circ g, \quad f : Y \rightarrow Y' \text{ and } g \in \text{Mor}_{\mathcal{C}}(X, Y). \end{aligned}$$

If we fix  $Y$  instead, then we obtain a contravariant *functor*  $R_Y : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$  *corepresented by*  $Y$ :

$$\begin{aligned} R^Y(X^\circ) &= \text{Mor}_{\mathcal{C}}(X, Y), \\ R^Y(f^\circ)(h) &= h \circ f, \quad f : X' \rightarrow X \text{ and } h \in \text{Mor}_{\mathcal{C}}(X, Y). \end{aligned}$$

**Definition 2.3.7** (Faithful / full functor). Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a functor. For each pair of objects  $X, Y \in \mathcal{C}$ , there is a set map  $\varphi_{X,Y} : \text{Mor}_{\mathcal{C}}(X, Y) \rightarrow \text{Mor}_{\mathcal{D}}(F(X), F(Y))$  given by  $\varphi_{X,Y}(f) = F(f)$ .

1.  $F$  is *faithful* if  $\varphi_{X,Y}$  is injective for all  $X, Y$ .

2.  $F$  is *full* if  $\varphi_{X,Y}$  is surjective for all  $X, Y$ .

**Example 2.3.8.** If  $\mathcal{C}' \subset \mathcal{C}$  is a subcategory, the inclusion  $\mathcal{C}' \hookrightarrow \mathcal{C}$  is faithful.

It is full if and only if  $\mathcal{C}'$  is a full subcategory.

**Definition 2.3.9** (Equivalence of categories). A functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  is an *equivalence* if

(i)  $F$  is full and faithful;

(ii) for any object  $Y \in \mathcal{D}$ , there exists  $X \in \mathcal{C}$  such that  $F(X) \cong Y$ .

**Proposition 2.3.10.** *Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a functor.*

1. *If  $F$  is full and faithful, then  $\mathcal{C}$  is equivalent to a full subcategory of  $\mathcal{D}$ .*
2. *If  $F$  is full and faithful, then  $X \cong Y$  in  $\mathcal{C}$  if and only if  $F(X) \cong F(Y)$  in  $\mathcal{D}$ .*
3. *If  $F$  is an equivalence, then  $F$  induces a bijection between isomorphism classes in  $\mathcal{C}$  and  $\mathcal{D}$ .*

**Example 2.3.11.** 1. Let  $\mathcal{C}' \subset \mathcal{C}$  be a full subcategory. Then  $\mathcal{C}' \hookrightarrow \mathcal{C}$  is an equivalence if and only if for every  $Y \in \mathcal{C}$ , there exists  $X \in \mathcal{C}'$  such that  $X \cong Y$ .

2. Consider  $\mathbf{Vect}_K$ , the category of vector spaces over  $K$ . The full subcategory of  $K$ -vector spaces of the form  $K^n$ , where  $n$  is any cardinal number, is equivalent to  $\mathbf{Vect}_K$ .

In particular, if we look at the full subcategory  $\mathbf{FdVect}_K$  of finite-dimensional vector spaces over  $K$ , it has as an equivalent full subcategory the vector spaces  $K^n$  for  $n \in \mathbb{N}$ , which is a small category.

## 2.4 Morphisms of functors

**Definition 2.4.1** (Morphisms of functors). Let  $F$  and  $G$  be two functors  $\mathcal{C} \rightarrow \mathcal{D}$ . A *morphism of functors*  $\alpha : F \rightarrow G$  (*natural transformation*) is a collection of morphisms  $\alpha_X : F(X) \rightarrow G(X)$  in  $\mathcal{D}$  for all objects  $X$  in  $\mathcal{C}$  such that for every morphism  $f : X \rightarrow Y$  in  $\mathcal{C}$ , the following diagram commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{\alpha_X} & G(X) \\ F(f) \downarrow & & \downarrow G(f) \\ F(Y) & \xrightarrow{\alpha_Y} & G(Y) \end{array}$$

**Definition 2.4.2** (Category of functors). Let  $\mathcal{C}$  and  $\mathcal{D}$  be two categories. The *category of functors from  $\mathcal{C}$  to  $\mathcal{D}$* , denoted  $\mathbf{Fun}(\mathcal{C}, \mathcal{D})$ , has objects the functors  $\mathcal{C} \rightarrow \mathcal{D}$  and morphisms the morphisms of functors.

**Notation.** Let  $\mathcal{C}, \mathcal{D}$  be categories and  $F, G : \mathcal{C} \rightarrow \mathcal{D}$  be functors. Write  $\text{Mor}_{\mathbf{Fun}}(F, G)$  for the collection of morphisms of functors  $F \rightarrow G$ .

**Proposition 2.4.3.** *If  $F, G : \mathcal{C} \rightarrow \mathcal{D}$  and  $\alpha : F \rightarrow G$  is a morphism of functors, then  $\alpha$  is an isomorphism in  $\mathbf{Fun}(\mathcal{C}, \mathcal{D})$  if and only if  $\alpha_X : F(X) \rightarrow G(X)$  is an isomorphism in  $\mathcal{D}$  for all  $X$  in  $\mathcal{C}$ .*

**Example 2.4.4.** 1. If  $\mathcal{I}$  is a small category, then the category  $\mathbf{Fun}(\mathcal{I}, \mathcal{D})$  is the category  $\text{Diag}_{\mathcal{I}}(\mathcal{D})$  of diagrams of shape  $\mathcal{I}$  in  $\mathcal{D}$ .

2. Let  $F : \mathbf{Grp} \rightarrow \mathbf{Grp}$  be the abelianization functor (with target  $\mathbf{Grp}$ ). For each  $G \in \mathbf{Grp}$ , set  $\alpha_G = \pi : G \rightarrow G/G'$ . Then  $\alpha$  is a morphism  $\text{id}_{\mathbf{Grp}} \rightarrow F$ .

3. Given  $f : X \rightarrow X'$  in a category  $\mathcal{C}$ , we have a morphism  $R^{X'}(Y) \rightarrow R^X(Y)$  for each  $Y$  given by  $g \mapsto g \circ f$ , and one can check that this produces a morphism of represented functors  $R^f : R^{X'} \rightarrow R^X$ . Hence there is a functor  $\mathcal{C}^{\text{op}} \rightarrow \mathbf{Fun}(\mathcal{C}, \mathbf{Set})$  with  $X^{\text{op}} \mapsto R^X$  and  $f^{\text{op}} \mapsto R^f$ .

Equivalently, we have a functor  $\mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathbf{Set}$  with  $(X^{\text{op}}, Y) \mapsto \text{Mor}_{\mathcal{C}}(X, Y)$ , or a functor  $\mathcal{C} \rightarrow \mathbf{Fun}(\mathcal{C}^{\text{op}}, \mathbf{Set})$  with  $Y \mapsto R_Y$ , where  $R_Y(X^{\text{op}}) = \text{Mor}_{\mathcal{C}}(X, Y)$ .

**Lemma 2.4.5** (Yoneda). *Let  $\mathcal{C}$  be a locally small category and fix  $X$  in  $\mathcal{C}$ . Let  $F : \mathcal{C} \rightarrow \mathbf{Set}$  be a functor. Then there is a bijection  $\varphi : \text{Mor}_{\mathbf{Fun}}(R^X, F) \rightarrow F(X)$  given by*

$$\varphi(\alpha) = \alpha_X(1_X).$$

*Proof.* Let  $\alpha : R^X \rightarrow F$  be a morphism of functors. By the very definition, for every morphism  $f : X \rightarrow Y$  in  $\mathcal{C}$  the diagram

$$\begin{array}{ccccc} R^X(X) & \xlongequal{\quad} & \text{Mor}(X, X) & \xrightarrow{\alpha_X} & F(X) \\ R^X(f) \downarrow & & \downarrow f \circ - & & \downarrow F(f) \\ R^X(Y) & \xlongequal{\quad} & \text{Mor}(X, Y) & \xrightarrow{\alpha_Y} & F(Y). \end{array}$$

is commutative. It follows that

$$\alpha_Y(f) = F(f)(\alpha_X(1_X)) = F(f)(\varphi(\alpha)).$$

We prove the injectivity of  $\varphi$ . Suppose  $\varphi(\alpha) = \varphi(\beta)$  for  $\alpha, \beta \in \text{Mor}_{\mathbf{Fun}}(R^X, F)$ . Then for every morphism  $f : X \rightarrow Y$  in  $\mathcal{C}$  we have

$$\alpha_Y(f) = F(f)(\varphi(\alpha)) = F(f)(\varphi(\beta)) = \beta_Y(f),$$

hence  $\alpha = \beta$ .

Next we prove the surjectivity of  $\varphi$ . Let  $u \in F(X)$ . For every object  $Y$  define a map

$$\alpha_Y : R^X(Y) = \text{Mor}(X, Y) \rightarrow F(Y)$$

by the formula

$$\alpha_Y(f) = F(f)(u).$$

Let  $g : Y \rightarrow Y'$  be a morphism. The diagram

$$\begin{array}{ccccc} R^X(Y) & \xlongequal{\quad} & \text{Mor}(X, Y) & \xrightarrow{\alpha_Y} & F(Y) \\ R^X(g) \downarrow & & \downarrow g \circ - & & \downarrow F(g) \\ R^X(Y') & \xlongequal{\quad} & \text{Mor}(X, Y') & \xrightarrow{\alpha_{Y'}} & F(Y'). \end{array}$$

is commutative. Indeed, for every  $f : X \rightarrow Y$ ,

$$(F(g) \circ \alpha_Y)(f) = F(g)(F(f)(u)) = F(g \circ f)(u) = \alpha_{Y'}(g \circ f)(u) = (\alpha_{Y'} \circ R^X(g))(f).$$

It follows that the collection  $(\alpha_Y)$  is a morphism of functors  $\alpha : R^X \rightarrow F$ . Finally,

$$\varphi(\alpha) = \alpha_X(1_X) = F(1_X)(u) = 1_{F(X)}(u) = u,$$

i.e.,  $\varphi$  is surjective. □

**Corollary 2.4.6.**  $\text{Mor}_{\mathbf{Fun}}(R^X, R^Y) \cong \text{Mor}_{\mathcal{C}}(Y, X)$ .

**Corollary 2.4.7.** Every natural transformation  $R^X \rightarrow R^Y$  is of the form  $R^f$  for a unique morphism  $f : Y \rightarrow X$ .

**Definition 2.4.8** (Presheaf of sets). A functor  $\mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$  is a *presheaf of sets*.

**Definition 2.4.9** (Representable functor). A functor  $F : \mathcal{C} \rightarrow \mathbf{Set}$  is *representable* if  $F$  is isomorphic to  $R^X$  for some  $X$ . We say that  $F$  is *represented by*  $X$ .

**Proposition 2.4.10.** Let  $F$  be representable and suppose  $F$  is represented by  $X$  and  $Y$ . Then there is a unique isomorphism  $f : X \rightarrow Y$  such that the following diagram commutes.

$$\begin{array}{ccc} R^Y & \xleftarrow{\text{iso}} & F \\ & \searrow R^f & \downarrow \text{iso} \\ & & R^X \end{array}$$

In other words, the object  $X$  is uniquely determined up to canonical isomorphism.

**Example 2.4.11.** 1.  $c_{\{*\}} : \mathcal{C} \rightarrow \mathbf{Set}$  is represented by any initial object of  $\mathcal{C}$ .

2. Let  $X$  be a set and define  $F : \mathbf{Grp}^{\text{op}} \rightarrow \mathbf{Set}$  by  $G^{\text{op}} \mapsto \{\text{left } G\text{-actions on } X\}$ . Then the symmetric group  $S(X)$  represents  $F$ .

3. Let  $\mathcal{C}$  be a locally small category with products and fix  $X, Y \in \mathcal{C}$ . Let  $F : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$  be given by  $F(Z) = \text{Mor}(Z, X) \times \text{Mor}(Z, Y)$ . This is corepresented by  $X \times Y$ , i.e.  $R_{X \times Y} \cong R_X \times R_Y$ . Similarly,  $R^X \times R^Y \cong R^{X * Y}$  if  $\mathcal{C}$  has coproducts.

4. Let  $X$  be an object of a category  $\mathcal{C}$  and consider the functor  $\mathbf{Fun}(\mathcal{C}, \mathbf{Set}) \rightarrow \mathbf{Set}$  given by  $F \mapsto F(X)$ . This functor is represented by  $R^X$ .

5. Let  $X$  be a set. The functor  $\mathbf{Grp} \rightarrow \mathbf{Set}$  taking a group  $G$  to the set of all maps  $X \rightarrow G$  is represented by the free group  $\text{Free}(X)$ .

6. The forgetful functor  $\mathbf{Grp} \rightarrow \mathbf{Set}$  is represented by  $\mathbb{Z}$ .

**Definition 2.4.12** (Adjunction). Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  and  $G : \mathcal{D} \rightarrow \mathcal{C}$  be two functors. There are two functors  $\mathcal{C}^{\text{op}} \times \mathcal{D} \rightarrow \mathbf{Set}$ , given by

$$(X^{\circ}, Y) \mapsto \text{Mor}_{\mathcal{C}}(X, G(Y)), \quad (X^{\circ}, Y) \mapsto \text{Mor}_{\mathcal{D}}(F(X), Y).$$

We say that  $F, G$  form an *adjunction pair* (or  $F$  and  $G$  are *adjoint*), with  $F$  a *left adjoint* to  $G$  and  $G$  a *right adjoint* to  $F$ , if these two functors are naturally isomorphic.

**Proposition 2.4.13.** Let  $F, G$  and  $F', G$  be adjunction pairs. Then  $F, F'$  are canonically isomorphic.

*Proof.* For any  $X$  in  $\mathcal{C}$ , the functor  $\mathcal{D} \rightarrow \mathbf{Set}$  given by

$$Y \mapsto \text{Mor}_{\mathcal{C}}(X, G(Y)) \cong \text{Mor}_{\mathcal{D}}(F(X), Y) = R^{F(X)}(Y)$$

is represented by  $F(X)$ . The same can be done for  $F'$ , so  $F(X) \cong F'(X)$  for all  $X$  in  $\mathcal{C}$ . By following the isomorphisms, we find that  $F$  and  $F'$  themselves are isomorphic.  $\square$

- Example 2.4.14.** 1. Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be an equivalence of categories and let  $G : \mathcal{D} \rightarrow \mathcal{C}$  be a *quasi-inverse* of  $F$ , i.e.,  $F \circ G$  is isomorphic to  $\text{id}_{\mathcal{D}}$  and  $G \circ F$  is isomorphic to  $\text{id}_{\mathcal{C}}$ . Then  $G$  is a left and right adjoint of  $F$ .
2. The product (resp., coproduct) functor  $\mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$  has a left (resp., right) adjoint functor  $X \mapsto (X, X)$ .
3. The forgetful functor  $\mathbf{Grp} \rightarrow \mathbf{Set}$  has as a left adjoint  $X \mapsto \text{Free}(X)$ .
4. The inclusion functor  $\mathbf{Ab} \hookrightarrow \mathbf{Grp}$  has as a left adjoint the abelianization functor.
5. Fix an integer  $n > 0$ . The functor  $F : \mathbf{Ab} \rightarrow \mathbf{Ab}$ ,  $F(A) = A[n] := \{a \in A : na = 0\}$ , has left adjoint  $B \mapsto B/nB$ .

**Definition 2.4.15** (Commuting with products). Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a functor between two categories with products. We say that  $F$  *commutes with products* if the unique natural morphism

$$F(\prod X_i) \rightarrow \prod F(X_i)$$

is an isomorphism for any family of objects  $X_i$  in  $\mathcal{C}$ .

**Proposition 2.4.16.** *If  $F : \mathcal{C} \rightarrow \mathcal{D}$  has a left adjoint, then  $F$  commutes with products.*

*Proof.* Let  $X_i$  be a family of objects in  $\mathcal{C}$  and  $Z$  in  $\mathcal{D}$ . Then

$$\begin{aligned} \text{Mor}_{\mathcal{D}}(Z, F(\prod X_i)) &\cong \text{Mor}_{\mathcal{C}}(G(Z), \prod X_i) \cong \prod \text{Mor}_{\mathcal{C}}(G(Z), X_i) \\ &\cong \prod \text{Mor}_{\mathcal{D}}(Z, F(X_i)) \cong \text{Mor}_{\mathcal{D}}(Z, \prod F(X_i)). \end{aligned} \quad \square$$

**Example 2.4.17.** The forgetful functor  $\mathbf{Grp} \rightarrow \mathbf{Set}$  commutes with products.

## 2.5 Limits and colimits

**Definition 2.5.1** (Limits / colimits). Let  $\mathcal{I}$  be a small category and  $X \in \mathcal{C}$ . Let  $c_X : \mathcal{I} \rightarrow \mathcal{C}$  be the constant functor and  $F : \mathcal{I} \rightarrow \mathcal{C}$  be some other functor. A morphism  $X \rightarrow Y$  induces a natural transformation  $c_X \rightarrow c_Y$ , so we have a functor  $\mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$  given by  $X^{\text{op}} \mapsto \text{Mor}_{\mathbf{Fun}}(c_X, F)$ . The *limit* of  $F$  is an object  $\lim F$  in  $\mathcal{C}$  corepresenting this functor. In other words, there are natural in  $X$  bijections

$$\text{Mor}_{\mathbf{Fun}}(c_X, F) \simeq \text{Mor}_{\mathcal{C}}(X, \lim F).$$

The *colimit* of  $F$  is an object  $\text{colim } F$  representing the functor  $\mathcal{C} \rightarrow \mathbf{Set}$  given by  $X \mapsto \text{Mor}_{\mathbf{Fun}}(F, c_X)$ . In other words, there are natural in  $X$  bijections

$$\text{Mor}_{\mathbf{Fun}}(F, c_X) \simeq \text{Mor}_{\mathcal{C}}(\text{colim } F, X).$$

**Definition 2.5.2** (Cone). Let  $\mathcal{I}$  be a small category with  $\text{Ob } \mathcal{I} = \{X_j\}_{j \in J}$ , let  $\mathcal{C}$  be another category, and  $F : \mathcal{I} \rightarrow \mathcal{C}$  be a functor. A *cone* of  $F$  is an object  $Y \in \mathcal{C}$  together with morphisms  $f_j : Y \rightarrow F(X_j)$  such that for any morphism  $g : X_j \rightarrow X_k$  in  $\mathcal{I}$ , we have  $F(g) \circ f_j = f_k$  in  $\mathcal{C}$ .

**Proposition 2.5.3** (Universal property of limits). *The limit of a diagram  $F : \mathcal{I} \rightarrow \mathcal{C}$  is specified by a terminal object in the category of cones to  $F$ .*

**Remark 2.5.4.** One can similarly construct co-cones by reversing all of the morphisms in  $\mathcal{C}$  in the definition of a cone, and then the colimit is an initial object in the category of co-cones of  $F$ .

**Example 2.5.5.** 1. If  $\mathcal{I}$  has no non-identity morphisms, then

$$\lim F = \prod_{i \in \mathcal{I}} F(i), \quad \text{colim } F = \coprod_{i \in \mathcal{I}} F(i).$$

2. If  $\mathcal{I}$  has a final (resp., initial) object  $i$ , then  $\text{colim } F = F(i)$  (resp.,  $\lim F = F(i)$ ).

3. Let  $\mathcal{I}$  be the following diagram and let  $\mathcal{C} = \mathbf{Set}$ .

$$\begin{array}{ccc} & & \cdot \\ & & \downarrow \\ \cdot & \longrightarrow & \cdot \end{array}$$

A functor  $F : \mathcal{I} \rightarrow \mathcal{C}$  is then a diagram of the following form in  $\mathcal{C}$ .

$$\begin{array}{ccc} & & A \\ & & \downarrow \\ B & \longrightarrow & C \end{array}$$

The limit of  $F$  is an object  $X$ , which has morphisms to  $A, B, C$ , such that for any other such object  $Y$ , there is a unique morphism  $f : Y \rightarrow X$  such that the following diagram commutes.

$$\begin{array}{ccccc} Y & & & & \\ & \searrow f & & \searrow & \\ & X & \longrightarrow & A & \\ & \downarrow & & \downarrow & \\ & B & \longrightarrow & C & \end{array}$$

The object  $X$  is called a *pullback* or *fiber product*, and is denoted  $A \times_C B$ . The colimit of the diagram is simply  $C$ .

4. Let  $\mathcal{I}$  be a small category. The functor  $\mathcal{C} \rightarrow \text{Diag}_{\mathcal{I}}(\mathcal{C})$  taking an object  $X$  to the constant diagram  $c_X$  has right adjoint  $F \mapsto \lim F$  and left adjoint  $F \mapsto \text{colim } F$ .
5. Let  $G$  be a group and let  $F : \underline{G} \rightarrow \mathcal{C}$  be given by a  $G$ -object  $X$  in  $\mathcal{C}$ . Then the object  $X^G$  of the " $G$ -fixed part of  $X$ " can be defined as  $\lim F$ . The "orbit space"  $X/G$  is defined as  $\text{colim } F$ .
6. The limit of the diagram

$$Y \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} X$$

is called the *equalizer* of  $f$  and  $g$ . It is an object  $Z$  together with a morphism  $i : Z \rightarrow Y$  such that  $f \circ i = g \circ i$  satisfying the following universal property: for every morphism  $j : U \rightarrow Y$

such that  $f \circ j = g \circ j$  there is a unique morphism  $k : U \rightarrow Z$  such that  $j = i \circ k$ . Equivalently, the sequence

$$\text{Maps}(U, Z) \xleftarrow{i} \text{Maps}(U, Y) \xrightleftharpoons[g]{f} \text{Maps}(U, X)$$

is an equalizer sequence of maps of sets. Dually one defines the *co-equalizer* of  $f$  and  $g$  as the colimit of the diagram.

Let  $\mathcal{I} \xrightarrow{F} \mathcal{C} \xrightarrow{G} \mathcal{D}$  be two functors. Applying  $G$  to the canonical morphism  $c_{\lim F} : \lim F \rightarrow F$ , we get a morphism  $c_{G(\lim F)} : G(\lim F) \rightarrow G \circ F$ , that yields a morphism  $\varphi : G(\lim F) \rightarrow \lim(G \circ F)$ . We say that a functor  $G : \mathcal{C} \rightarrow \mathcal{D}$  *commutes with limits* if for every functor  $F : \mathcal{I} \rightarrow \mathcal{C}$  the morphism  $\varphi$  is an isomorphism.

**Proposition 2.5.6.** *If a functor  $G : \mathcal{C} \rightarrow \mathcal{D}$  has a left (resp., right) adjoint, then  $G$  commutes with limits (resp., colimits).*

## 2.6 Additive and abelian categories

**Definition 2.6.1** (Preadditive category). A locally small category  $\mathcal{A}$  is called *preadditive* if for every two objects  $X$  and  $Y$  in  $\mathcal{A}$  the set  $\text{Mor}_{\mathcal{A}}(X, Y)$  is equipped with the structure of an abelian group (written additively) such that the composition is bilinear, i.e.,  $(f + f') \circ g = f \circ g + f' \circ g$  and  $f \circ (g + g') = f \circ g + f \circ g'$ .

If  $\mathcal{A}$  is a preadditive category, then so is the opposite category  $\mathcal{A}^{\text{op}}$ .

**Definition 2.6.2** (Biproduct). Let  $\mathcal{A}$  be a preadditive category. Let  $X$  and  $Y$  be two objects in  $\mathcal{A}$ . A *biproduct* of  $X$  and  $Y$  is an object  $Z$  together with the four morphisms  $i : X \rightarrow Z$ ,  $p : Z \rightarrow X$ ,  $j : Y \rightarrow Z$  and  $q : Z \rightarrow Y$  such that  $p \circ i = 1_X$ ,  $q \circ i = 0$ ,  $p \circ j = 0$ ,  $q \circ j = 1_Y$  and  $i \circ p + j \circ q = 1_Z$ .

A biproduct of  $X$  and  $Y$  in  $\mathcal{A}$  yields a biproduct of  $X^{\text{op}}$  and  $Y^{\text{op}}$  in  $\mathcal{A}^{\text{op}}$ .

Let  $(Z, i, j, p, q)$  be a biproduct of  $X$  and  $Y$ . Then  $(Z, p, q)$  is a product of  $X$  and  $Y$ . Indeed, if  $f : U \rightarrow X$  and  $g : U \rightarrow Y$  be two morphisms, the setting  $h = i \circ f + j \circ g : U \rightarrow Z$ , we have  $p \circ h = f$  and  $q \circ h = g$ . Moreover if  $h' : U \rightarrow Z$  satisfies  $p \circ h' = f$  and  $q \circ h' = g$ , then  $h' = (i \circ p + j \circ q) \circ h' = i \circ f + j \circ g = h$ .

Similarly,  $(Z, i, j)$  is a coproduct of  $X$  and  $Y$ . Thus, if  $X$  and  $Y$  admit a biproduct  $Z$ , then  $Z$  is also a product and coproduct of  $X$  and  $Y$ .

Let  $(Z, p, q)$  be a product of two objects  $X$  and  $Y$ . We claim that it extends to a biproduct of  $X$  and  $Y$ . Indeed, we let  $i : X \rightarrow Z$  be the morphism determined by the pair of morphisms  $1_X : X \rightarrow X$  and  $0 : X \rightarrow Y$ . Similarly,  $j : Y \rightarrow Z$  is determined by the pair of morphisms  $0 : Y \rightarrow X$  and  $1_Y : Y \rightarrow Y$ . This proves the claim.

Equivalently a coproduct  $(Z, i, j)$  of two objects  $X$  and  $Y$  extends naturally to a biproduct of  $X$  and  $Y$ . Thus, to give a biproduct of two objects in a preadditive category is equivalent to give their product (or coproduct).

**Definition 2.6.3** (Additive category). A preadditive category  $\mathcal{A}$  is called *additive* if finite products exist in  $\mathcal{A}$ .

Note that the product, coproduct and biproduct of every two objects in an additive category coincide. In particular, initial and final objects coincide. They are called *zero objects*. A zero object  $X$  is characterized by the property  $1_X = 0_X$ .

- Example 2.6.4.** 1. The category **Ab** of abelian groups is additive.
2. The category of morphisms in **Ab** is additive.
3. The category of short exact sequences in **Ab** is additive.
4. A full subcategory of an additive category that is closed under finite products is additive. For example, the category of free groups  $\mathbb{Z}^n$  is additive.
5. If  $\mathcal{A}$  is an additive category, then the category of functors  $\mathcal{A}^{\text{op}} \rightarrow \mathbf{Ab}$  (the category of presheaves of abelian groups on  $\mathcal{A}$ ) is additive.

A functor  $F : \mathcal{A} \rightarrow \mathcal{B}$  between additive categories is called *additive* if  $F(g + h) = F(g) + F(h)$  for every two morphisms  $g, h : X \rightarrow Y$  in  $\mathcal{A}$ . If  $F$  is additive, then  $F(0_X) = 0_{F(X)}$  for every object  $X$  in  $\mathcal{A}$ . It follows that for a zero object  $0$  in  $\mathcal{A}$  we have  $1_{F(0)} = F(1_0) = F(0_0) = 0_{F(0)}$ , i.e.,  $F(0)$  is a zero object in  $\mathcal{B}$ . Thus, an additive functors take zero objects to zero objects.

**Example 2.6.5.** 1. Let  $\mathcal{A}$  be an additive category and  $A$  an object in  $\mathcal{A}$ . The functor  $R^A : \mathcal{A} \rightarrow \mathbf{Ab}$  taking an object  $X$  to the group  $\text{Mor}_{\mathcal{A}}(A, X)$  is additive. The functor  $R^A$  is called *represented by  $A$* . An additive functor  $F : \mathcal{A} \rightarrow \mathbf{Ab}$  is called *representable* if  $F$  is isomorphic to  $R^A$  for some  $A$ . By an additive analog of the Yoneda Lemma, the object  $A$  is uniquely determined up to canonical isomorphism. In a similar fashion one defines *corepresentable* functors  $\mathcal{A}^{\text{op}} \rightarrow \mathbf{Ab}$ .

2. The constant functor  $\mathcal{A} \rightarrow \mathbf{Ab}$  taking any object to a fixed abelian group  $A$  is not additive unless  $A$  is zero.

Let  $(Z, i, j, p, q)$  be a biproduct in  $\mathcal{A}$ . Then for an additive functor  $F : \mathcal{A} \rightarrow \mathcal{B}$ , the tuple  $(F(Z), F(i), F(j), F(p), F(q))$  is a biproduct in  $\mathcal{B}$ . It follows that  $F$  takes direct products (coproducts) to direct products (coproducts).

Let  $\mathcal{A}$  be an additive category and  $f : A \rightarrow B$  a morphism in  $\mathcal{A}$ . The *kernel of  $f$* , denoted  $\text{Ker}(f)$  is the equalizer of

$$A \xrightarrow[f]{f} B.$$

By definition  $\text{Ker}(f)$  is equipped with a morphism  $i : \text{Ker}(f) \rightarrow A$  such that for every object  $X$  the sequence

$$0 \rightarrow \text{Mor}(X, \text{Ker}(f)) \xrightarrow{i_*} \text{Mor}(X, A) \xrightarrow{f_*} \text{Mor}(X, B) \quad (1)$$

is exact. In different words, the kernel of  $f$  corepresents the functor  $\text{Ker}(R_A \xrightarrow{f_*} R_B)$ .

Similarly, the *cokernel of  $f$* , denoted  $\text{Coker}(f)$  is the co-equalizer of the pair  $f$  and  $0$ . By definition  $\text{Coker}(f)$  is equipped with a morphism  $j : B \rightarrow \text{Coker}(f)$  such that for every object  $Y$  the sequence

$$0 \rightarrow \text{Mor}(\text{Coker}(f), Y) \xrightarrow{j^*} \text{Mor}(B, Y) \xrightarrow{f^*} \text{Mor}(A, Y)$$

is exact. The cokernel of  $f$  represents the functor  $\text{Ker}(R^B \xrightarrow{f^*} R^A)$ .

We also define the *image*  $\text{Im}(f)$  by the formula

$$\text{Im}(f) = \text{Ker}(B \xrightarrow{j} \text{Coker}(f))$$

and the *coimage*  $\text{Coim}(f)$  by

$$\text{Coim}(f) = \text{Coker}(\text{Ker}(f) \xrightarrow{i} A).$$

**Definition 2.6.6** (Preabelian category). An additive category  $\mathcal{A}$  is called *pre-abelian* if every morphism in  $\mathcal{A}$  has kernel and cokernel (and hence image and coimage).

**Example 2.6.7.** 1. The category **Ab** of abelian groups is pre-abelian, but the full subcategory of free groups is not preabelian.  
2. The category of pairs  $(A, B)$  of abelian groups such that  $A$  is a subgroup of  $B$  and morphisms  $(A', B') \rightarrow (A, B)$  being group homomorphisms  $f : B' \rightarrow B$  such that  $f(A') \subset A$  is pre-abelian.

Consider a commutative square in a pre-abelian category:

$$\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ \downarrow & & \downarrow \\ A & \xrightarrow{f} & B. \end{array}$$

The composition  $\text{Ker}(f') \rightarrow A' \rightarrow A \xrightarrow{j} B$  coincides with the composition  $\text{Ker}(f') \rightarrow A' \xrightarrow{j'} B' \rightarrow B$  and hence is zero. By the definition of the kernel (and similarly for the cokernel), there are unique morphisms  $\text{Ker}(f') \rightarrow \text{Ker}(f)$  and  $\text{Coker}(f') \rightarrow \text{Coker}(f)$  making the diagram

$$\begin{array}{ccccccc} \text{Ker}(f') & \longrightarrow & A' & \xrightarrow{f'} & B' & \longrightarrow & \text{Coker}(f') \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \text{Ker}(f) & \longrightarrow & A & \xrightarrow{f} & B & \longrightarrow & \text{Coker}(f) \end{array}$$

commutative. Moreover, we can view the kernels and cokernels as functors  $\text{Arr}(\mathcal{A}) \rightarrow \mathcal{A}$ .

Let

$$A \xrightarrow{f} B \xrightarrow{g} C \tag{2}$$

two morphisms in a pre-abelian category such that  $g \circ f = 0$ . By the definition of the cokernel, there is a unique morphism  $\underline{k} : C \rightarrow C$  such that  $g$  is the composition  $B \xrightarrow{j} \text{Coker}(f) \xrightarrow{k} C$ . The commutative square

$$\begin{array}{ccc} B & \xrightarrow{j} & \text{Coker}(f) \\ \downarrow 1_B & & \downarrow k \\ B & \xrightarrow{g} & C \end{array}$$

yields a morphism

$$\mathrm{Im}(f) = \mathrm{Ker}(j) \rightarrow \mathrm{Ker}(g).$$

We say that the sequence (2) with  $g \circ f = 0$  is *exact* if the morphism  $\mathrm{Im}(f) \rightarrow \mathrm{Ker}(g)$  is an isomorphism.

**Lemma 2.6.8.** *Let  $f : A \rightarrow B$  be a morphism in a pre-abelian category. TFAE:*

- (1) *The sequence  $0 \rightarrow A \xrightarrow{f} B$  is exact;*
- (2)  *$\mathrm{Ker}(f) = 0$ ;*
- (3)  *$f$  is a monomorphism, i.e., the map  $f_* : \mathrm{Mor}(X, A) \rightarrow \mathrm{Mor}(X, B)$  is injective for any object  $X$ .*

*Proof.* (1)  $\Leftrightarrow$  (2): Clearly, the image of  $0 \rightarrow A$  is 0. Thus the sequence is exact if and only if  $\mathrm{Ker}(f) = 0$ .

(2)  $\Leftrightarrow$  (3): In view of (1),  $f_*$  is injective for all  $X$  if and only if  $\mathrm{Mor}(X, \mathrm{Ker}(f)) = 0$  for all  $X$ . The latter is equivalent to  $\mathrm{Ker}(f) = 0$ .  $\square$

Dually, the sequence  $0A \xrightarrow{f} B \rightarrow 0$  is exact if and only if  $\mathrm{Coker}(f) = 0$  if and only if  $f$  is an epimorphism.

Let  $f : A \rightarrow B$  be a morphism in a pre-abelian category  $\mathcal{A}$ . Note that we have canonical morphisms  $k : A \rightarrow \mathrm{Coim}(f)$  and  $l : \mathrm{Im}(f) \rightarrow B$ . Since the composition  $A \xrightarrow{f} B \xrightarrow{j} \mathrm{Coker}(f)$  is zero, by the definition of the image of  $f$ , there is a unique morphism  $m : A \rightarrow \mathrm{Im}(f)$  such that  $l \circ m = f$ . As the composition  $\mathrm{Ker}(f) \xrightarrow{i} A \xrightarrow{f} B$  is zero and the map

$$l_* : \mathrm{Mor}(\mathrm{Ker}(f), \mathrm{Im}(f)) \rightarrow \mathrm{Mor}(\mathrm{Ker}(f), B)$$

is injective (since  $\mathrm{Im}(f)$  is the kernel!), the composition  $\mathrm{Ker}(f) \xrightarrow{i} A \xrightarrow{m} \mathrm{Im}(f)$  is zero. By the definition of  $\mathrm{Coim}(f)$ , there is a unique morphism  $g : \mathrm{Coim}(f) \rightarrow \mathrm{Im}(f)$  such that  $g \circ k = m$ . Since  $l \circ g \circ k = l \circ m = f$ , the morphism  $f$  factors into the composition

$$A \xrightarrow{k} \mathrm{Coim}(f) \xrightarrow{g} \mathrm{Im}(f) \xrightarrow{l} B.$$

**Example 2.6.9.** Let  $f : A \rightarrow B$  be a morphism in  $\mathbf{Ab}$ . Then  $\mathrm{Coker}(f) = B/\mathrm{Im}(f)$  and  $\mathrm{Coim}(f) = A/\mathrm{Ker}(f)$ . The morphism  $g : A/\mathrm{Ker}(f) \rightarrow \mathrm{Im}(f)$  is the isomorphism given by the First Isomorphism Theorem.

**Definition 2.6.10** (Abelian category). A pre-abelian category  $\mathcal{A}$  is called *abelian* if for every morphism  $f : A \rightarrow B$ , the induced morphism  $g : \mathrm{Coim}(f) \rightarrow \mathrm{Im}(f)$  is an isomorphism, i.e., the First Isomorphism Theorem holds in  $\mathcal{A}$ .

**Example 2.6.11.** 1. The category  $\mathbf{Ab}$  of abelian groups is abelian.

2. The full subcategory of finite groups in  $\mathbf{Ab}$  is abelian.

3. If  $\mathcal{A}$  is an abelian category, so is  $\mathcal{A}^{\mathrm{op}}$ .

4. For every category  $\mathcal{C}$  and an abelian category  $\mathcal{A}$ , the category of functors  $\mathcal{C} \rightarrow \mathcal{A}$  is abelian. In particular, the category of diagrams of a given shape in an abelian category is abelian.

5. The category of pairs of abelian groups as above is not abelian. Let  $A' \subset A \subset B$  be subgroups and let  $f : (A', B) \rightarrow (A, B)$  be the morphism given by the identity of  $B$ . We have  $\text{Ker}(f) = 0 = \text{Coker}(f)$ ,  $\text{Coim}(f) = (A', B)$  and  $\text{Im}(f) = (A, B)$ . The morphism  $g = f : (A', B) \rightarrow (A, B)$  is not an isomorphism if  $A' \neq A$ .

**Lemma 2.6.12.** *If  $f : A \rightarrow B$  be a monomorphism in an abelian category, then  $\text{Im}(f) = A$ . Dually, if  $g : B \rightarrow C$  be an epimorphism, then  $\text{Coim}(g) = C$ .*

*Proof.* By Lemma 2.6.8,  $\text{Ker}(f) = 0$ , hence

$$\text{Coim}(f) = \text{Coker}(\text{Ker}(f) \rightarrow A) = \text{Coker}(0 \rightarrow A) = A.$$

Since the category is abelian,  $\text{Im}(f) = \text{Coim}(f) = A$ . □

**Corollary 2.6.13.** *Let  $f : A \rightarrow B$  be a morphism in an abelian category. TFAE:*

1.  $f$  is an isomorphism;
2.  $\text{Ker}(f) = 0 = \text{Coker}(f)$ ;
3.  $f$  is a monomorphism and epimorphism.

*Proof.* (1)  $\Rightarrow$  (2) is clear and (2)  $\Leftrightarrow$  (3) is proved in Lemma 2.6.8.

(3)  $\Rightarrow$  (1): By Lemma 2.6.12,  $f$  is isomorphic to the isomorphism  $\text{Coim}(f) \xrightarrow{\sim} \text{Im}(f)$ . □

**Proposition 2.6.14.** *Let  $\mathcal{A}$  be an abelian category. Then*

- 1) A sequence  $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C$  in  $\mathcal{A}$  is exact if and only if  $A = \text{Ker}(g)$ .
- 2) A sequence  $A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$  in  $\mathcal{A}$  is exact if and only if  $C = \text{Coker}(f)$ .

*Proof.* We prove 1. If the sequence is exact, then  $f$  is a monomorphism by Lemma 2.6.8. Hence  $A = \text{Im}(f) = \text{Ker}(g)$  by Lemma 2.6.12. Conversely, since  $A = \text{Ker}(g)$ , the morphism  $f$  is a monomorphism in view of (1). By Lemma 2.6.12 again,  $A = \text{Im}(f)$ , hence  $\text{Im}(f) = \text{Ker}(g)$ , i.e., the sequence is exact. □

**Corollary 2.6.15.** *Let  $\mathcal{A}$  be an abelian category. Then*

- 1) A sequence  $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C$  in  $\mathcal{A}$  is exact if and only if the sequence of abelian groups

$$0 \rightarrow \text{Mor}(X, A) \xrightarrow{f_*} \text{Mor}(X, B) \xrightarrow{g_*} \text{Mor}(X, C)$$

is exact for every  $X$ .

- 2) A sequence  $A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$  in  $\mathcal{A}$  is exact if and only if the sequence of abelian groups

$$0 \rightarrow \text{Mor}(C, Y) \xrightarrow{g^*} \text{Mor}(B, Y) \xrightarrow{f^*} \text{Mor}(A, Y)$$

is exact for every  $Y$ .

**Example 2.6.16.** For a morphism  $f : A \rightarrow B$  in an abelian category the sequence

$$0 \rightarrow \operatorname{Ker}(f) \rightarrow A \xrightarrow{f} B \rightarrow \operatorname{Coker}(f) \rightarrow 0$$

is exact.

An additive functor  $F : \mathcal{A} \rightarrow \mathcal{B}$  between two abelian categories is called *left exact* if for every exact sequence  $0 \rightarrow A \rightarrow B \rightarrow C$  in  $\mathcal{A}$  the sequence

$$0 \rightarrow F(A) \rightarrow F(B) \rightarrow F(C)$$

is exact in  $\mathcal{B}$ . Similarly,  $F$  is called *left exact* if for every exact sequence  $A \rightarrow B \rightarrow C \rightarrow 0$  in  $\mathcal{A}$  the sequence

$$F(A) \rightarrow F(B) \rightarrow F(C) \rightarrow 0$$

is exact in  $\mathcal{B}$ . The functor  $F$  is called *exact* if it is left and right exact.

**Example 2.6.17.** For an object  $A$  of an abelian category  $\mathcal{A}$  the represented functor  $R^A : \mathcal{A} \rightarrow \mathbf{Ab}$  is left exact. The corepresented functor  $R_A : \mathcal{A}^{\text{op}} \rightarrow \mathbf{Ab}$  is also left exact.

**Theorem 2.6.18** (Mitchell). *Let  $\mathcal{A}$  be a small abelian category. Then there is a ring  $R$  and a full faithful exact functor from  $\mathcal{A}$  to the abelian category of left  $R$ -modules.*

A commutative diagram

$$\begin{array}{ccccccc} A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & C \\ \downarrow f & & \downarrow g & & \downarrow h & & \\ 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & C' \end{array}$$

in an abelian category with exact rows yields a sequence

$$\operatorname{Ker}(f) \rightarrow \operatorname{Ker}(g) \rightarrow \operatorname{Ker}(h) \rightarrow \operatorname{Coker}(f) \rightarrow \operatorname{Coker}(g) \rightarrow \operatorname{Coker}(h)$$

**Theorem 2.6.19** (Snake Lemma). *This sequence is exact.*

**Theorem 2.6.20.** *Let  $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$  be a short exact sequence in an abelian category. Then the following are equivalent:*

- (1) (right split) there exists  $g' : C \rightarrow B$  such that  $g \circ g' = \operatorname{id}_C$ ;
- (2) (left split) there exists  $f' : B \rightarrow A$  such that  $f' \circ f = \operatorname{id}_A$ ;
- (3) this short exact sequence is isomorphic to  $0 \rightarrow A \rightarrow A \oplus C \rightarrow C \rightarrow 0$ .

*Proof.* (2)  $\implies$  (3) Given  $f'$  as in (2), we have a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{g} & C \longrightarrow 0 \\ & & \downarrow 1_A & & \downarrow (f', g) & & \downarrow 1_C \\ 0 & \longrightarrow & A & \longrightarrow & A \oplus C & \longrightarrow & C \longrightarrow 0 \end{array}$$

By the Snake Lemma, the middle morphism is an isomorphism.

(1)  $\implies$  (3) is similar.

(3)  $\implies$  (1) and (3)  $\implies$  (2) are obvious.  $\square$

**Definition 2.6.21.** A short exact sequence  $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$  in an abelian category is called *split* if all conditions in the theorem hold.

**Proposition 2.6.22.** Let  $P$  be an object of an abelian category. The following are equivalent:

- (1) the functor  $\text{Mor}(P, -)$  is exact;
- (2) for every diagram of the form below, there exists  $r : P \rightarrow B$  such that  $g \circ r = h$ ;

$$\begin{array}{ccc} & & P \\ & \swarrow r & \downarrow h \\ B & \xrightarrow{g} & C \end{array}$$

- (3) every short exact sequence  $0 \rightarrow N \rightarrow M \rightarrow P \rightarrow 0$  is split.

*Proof.* (1)  $\iff$  (2) Let  $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$  be a short exact sequence. Then

$$0 \longrightarrow \text{Mor}(P, A) \xrightarrow{f_*} \text{Mor}(P, B) \xrightarrow{g_*} \text{Mor}(P, C) \longrightarrow 0$$

is exact if and only if  $g_*$  is surjective, or equivalently, for all  $h \in \text{Mor}(P, C)$ , there exists  $r \in \text{Mor}(P, B)$  such that  $g_*(r) = g \circ r = h$ .

- (2)  $\implies$  (3) Given such a short exact sequence, we construct the diagram

$$\begin{array}{ccccccc} & & & & P & & \\ & & & \swarrow r & \downarrow 1_P & & \\ 0 & \longrightarrow & N & \longrightarrow & M & \xrightarrow{s} & P \longrightarrow 0 \end{array}$$

Then  $s \circ r = 1_P$ , so  $s$  gives the required splitting.

- (3)  $\implies$  (2) Suppose we have such a diagram, which equivalently is a diagram of the below form with  $B \rightarrow C \rightarrow 0$  exact.

$$\begin{array}{ccc} & & P \\ & & \downarrow h \\ B & \xrightarrow{g} & C \longrightarrow 0 \end{array}$$

Let

$$A := \text{Ker}(g), \quad D := \text{Ker}(B \oplus P \xrightarrow{(g,h)} C).$$

We have a commutative diagram with the exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{g} & C \longrightarrow 0 \\ & & \downarrow k & & \downarrow (1_B, 0) & & \downarrow 1_C \\ 0 & \longrightarrow & D & \xrightarrow{(i,j)} & B \oplus P & \xrightarrow{(g,h)} & C \longrightarrow 0. \end{array}$$

By the Snake Lemma the top row of the commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \xrightarrow{k} & D & \xrightarrow{-j} & P \longrightarrow 0 \\ & & \downarrow 1_A & & \downarrow i & & \downarrow h \\ 0 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{g} & C \longrightarrow 0. \end{array}$$

is exact. The upper row is split, so there exists  $l : P \rightarrow D$  with  $(-j) \circ l = 1_P$ . Take

$$r = i \circ l : P \rightarrow B.$$

Then

$$g \circ r = g \circ i \circ l = h \circ (-j) \circ l = h \circ 1_P = h. \quad \square$$

**Definition 2.6.23.** An object  $P$  is called *projective* if  $P$  satisfies all equivalent conditions of the proposition.

Dually, we get the following statement.

**Proposition 2.6.24.** *Let  $Q$  be an object of an abelian category. The following are equivalent:*

- (1) *the functor  $\text{Mor}(-, Q)$  is exact;*
- (2) *for every diagram of the form below, there exists  $r : B \rightarrow Q$  such that  $r \circ f = h$ ;*

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow h & \swarrow \text{---} r & \\ Q & & \end{array}$$

- (3) *every short exact sequence  $0 \rightarrow Q \rightarrow M \rightarrow N \rightarrow 0$  is split.*

**Definition 2.6.25.** An object  $Q$  is called *injective* if  $Q$  satisfies all equivalent conditions of the proposition.